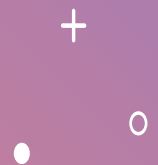


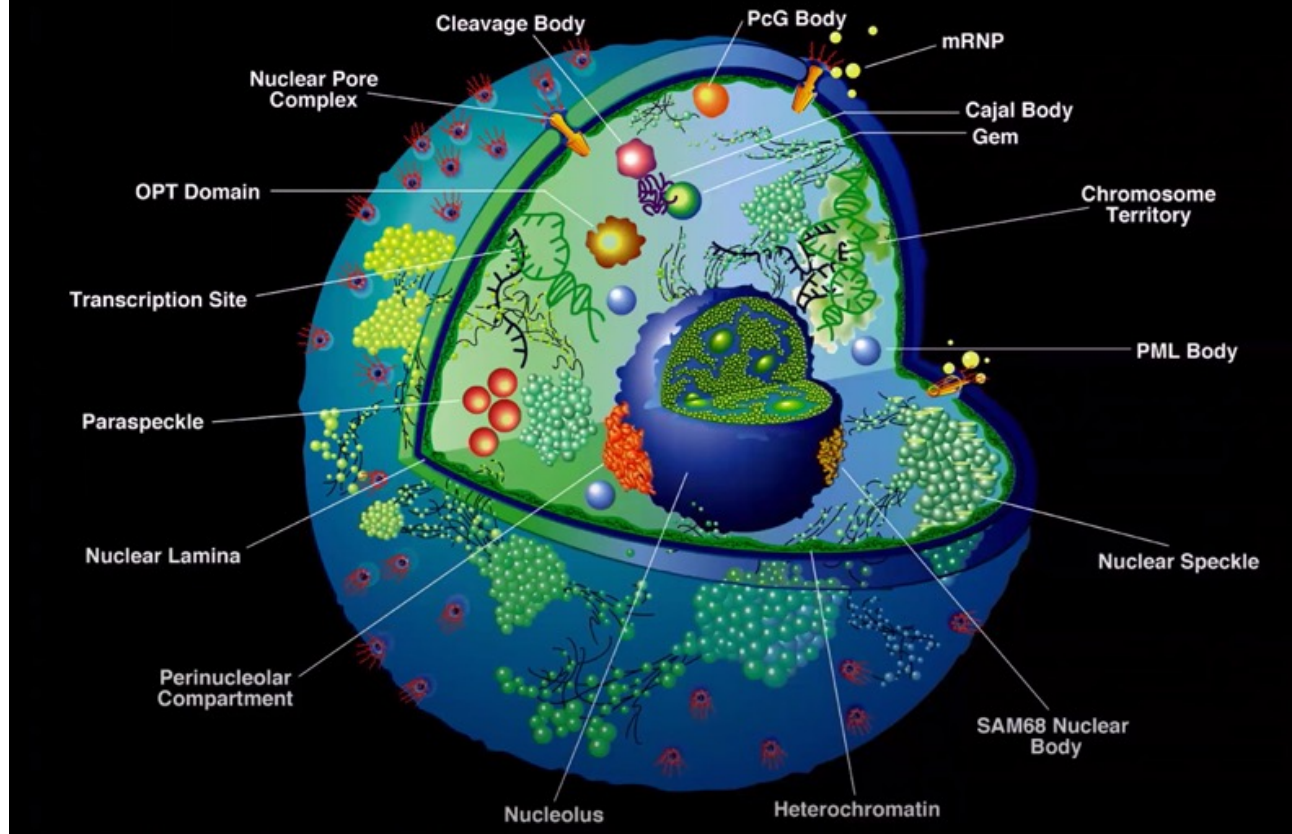


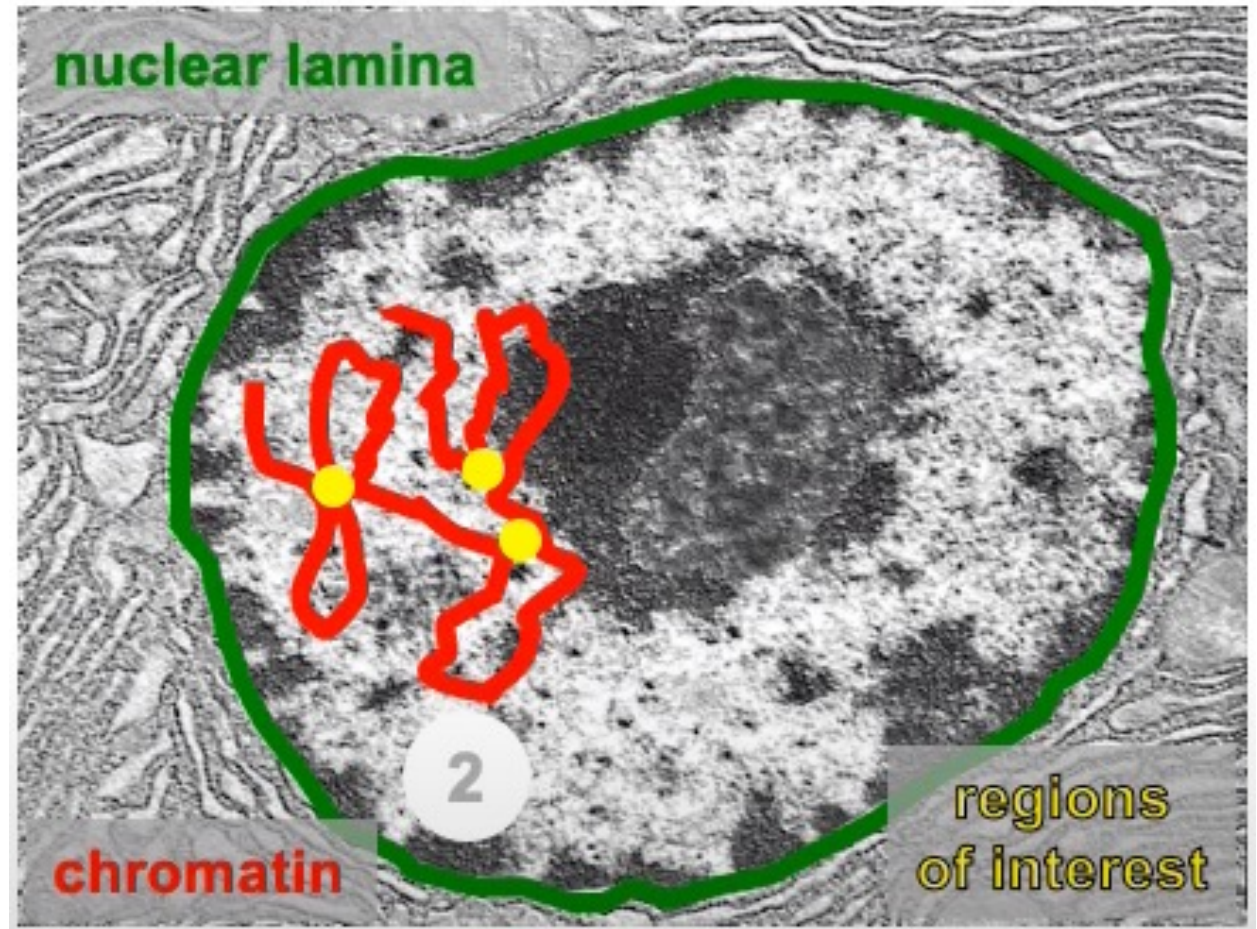
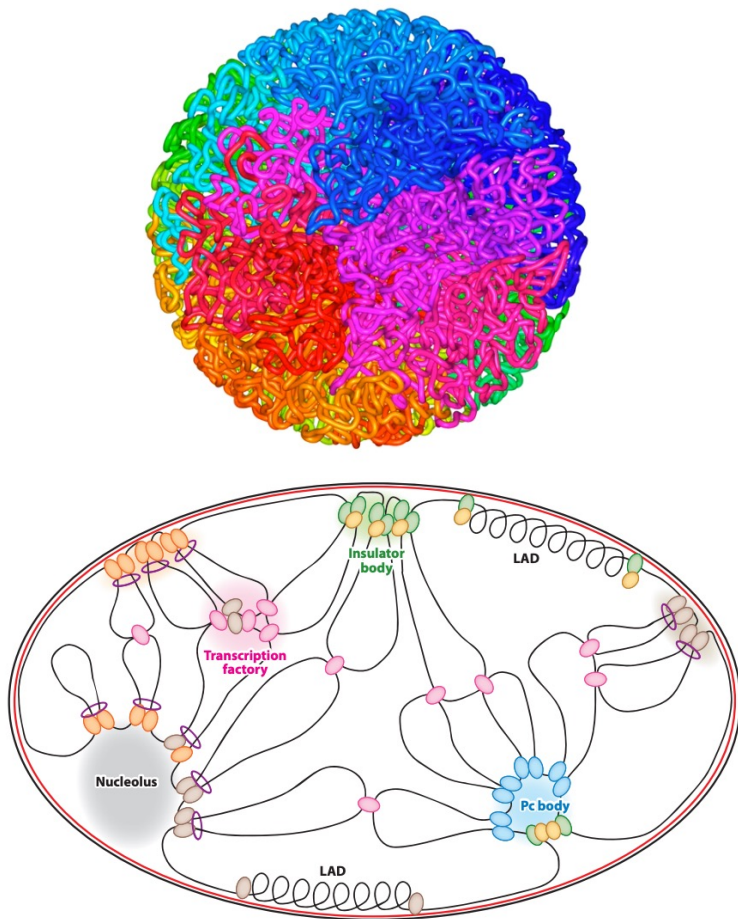
Genome and Transcriptome in Cancer (Integration of omics data)

Chapter22-23

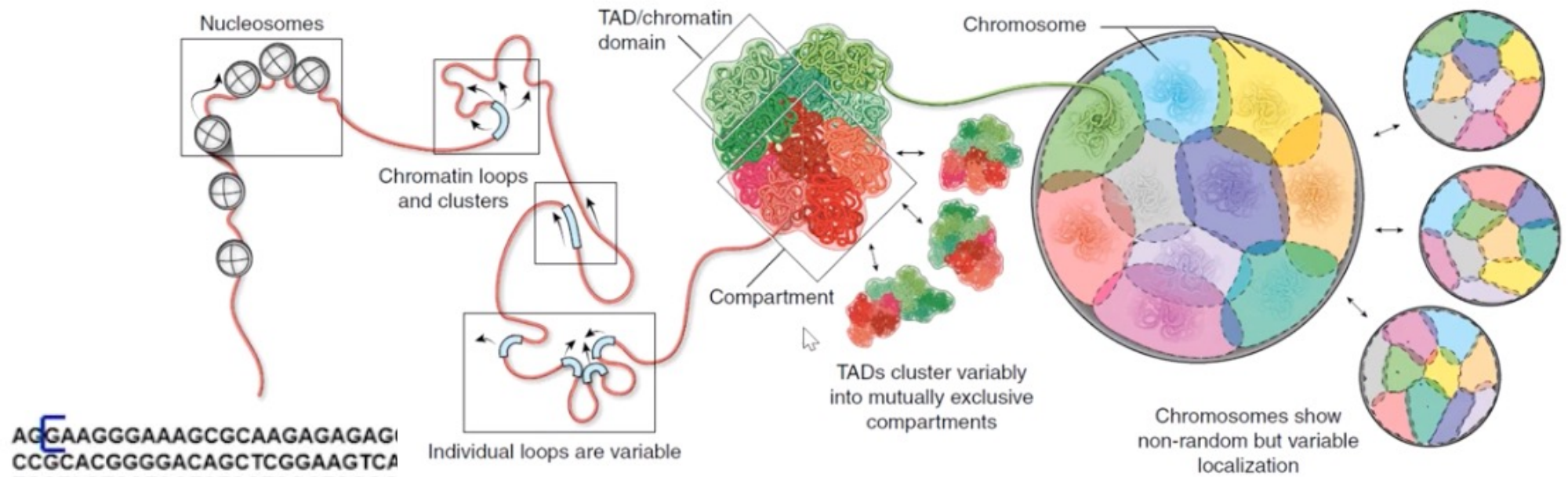


A model of the (mammalian) nucleus

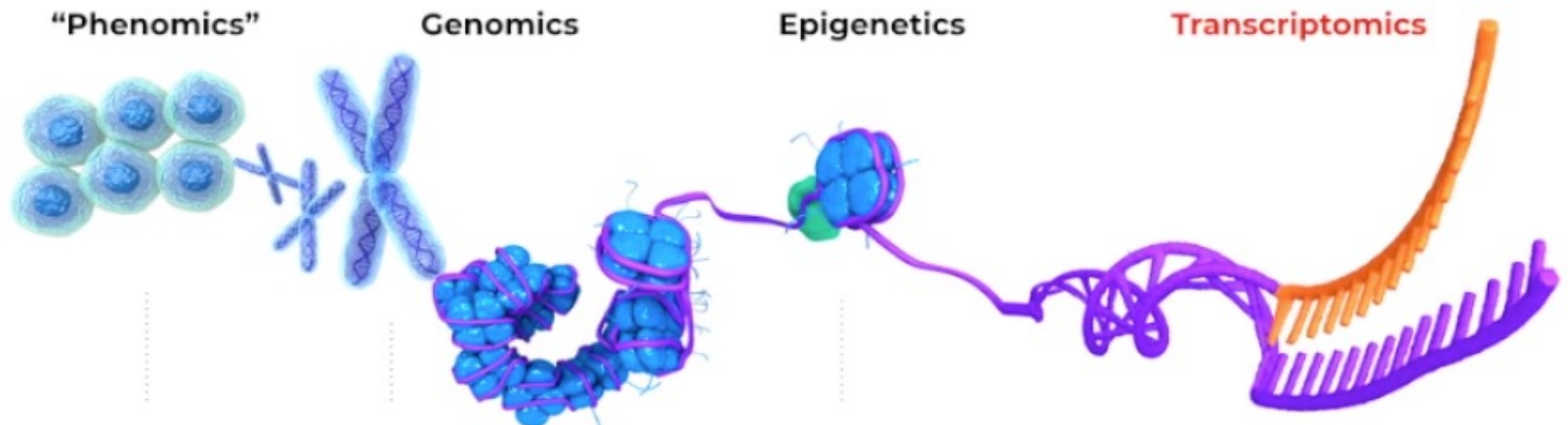




Van Bortle K, Corces VG, Nature Reviews Genetics, 2012



ACGAAGGGAAAGCGCAAGAGAGAGI
 CCGCACGGGGACAGCTCGGAAGTCA
 TGCTAGTCCTCGTCTCCTCGCTGCGG
 GAAGAGGAGGCACCCCAAAAAGCTG
 GAAGACCCTAGGCGCCAGCGGAAGC
 AACTCACCCCAATTACAACCCCGAC
 GATGACTCAGTAGGAACCCAGCGCC
 CCGAACACTTCCAGACTTAACTCTGT
 CCACCCACCTGCCCGCCCCCATTCT
 GAGAAGTTAGGGTGTGGTGGGGGTG



Clinical symptoms of Leukemia

Symptoms Of Leukemia

Swollen lymph nodes



Nose bleeds



Bleeding easily



Night sweats



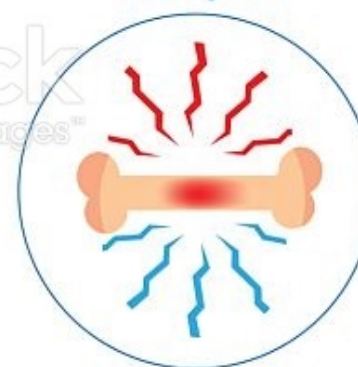
Weight loss



Fatigue and Loss of appetite



Bone pain



Severe infections



Fever



Chills

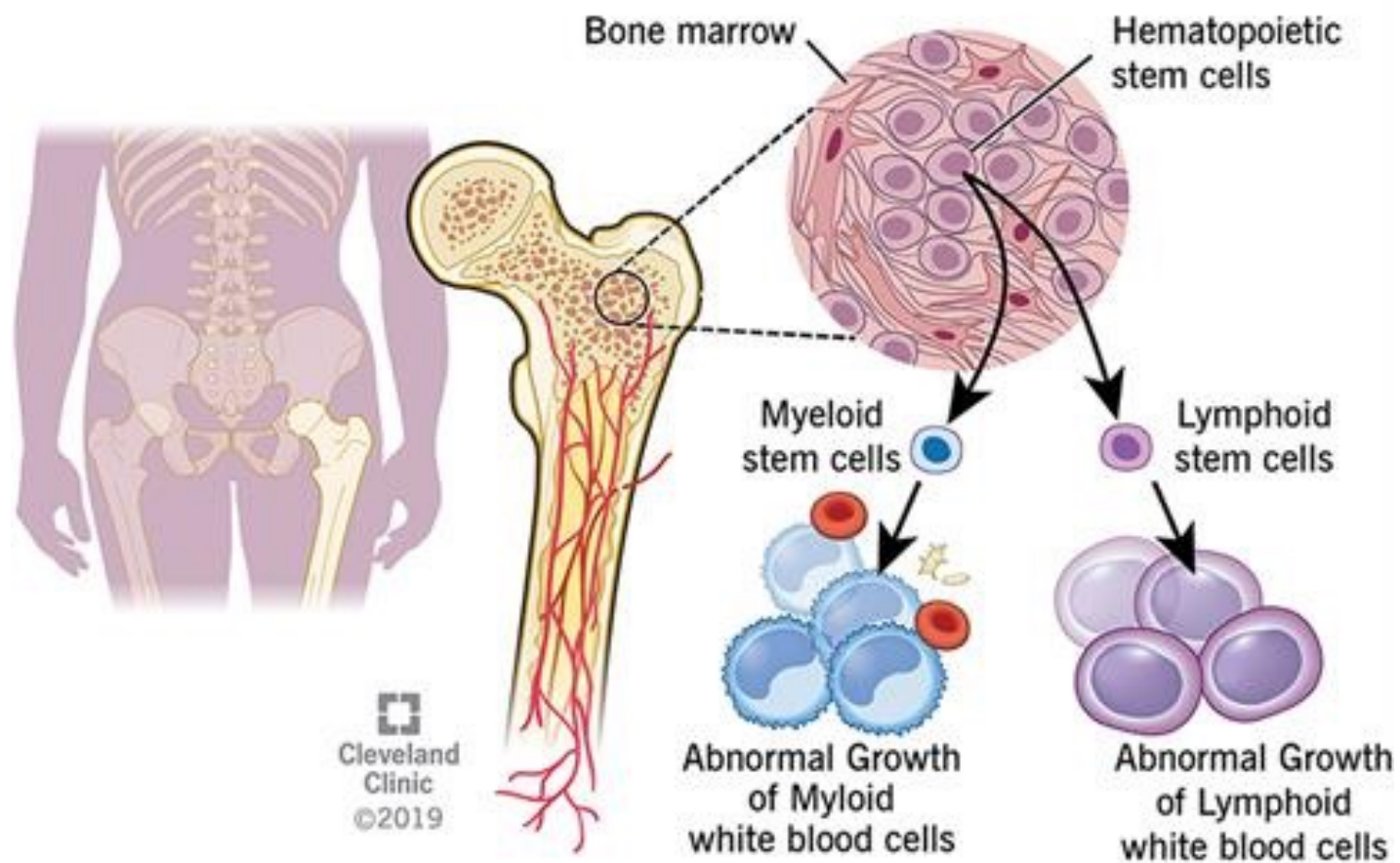


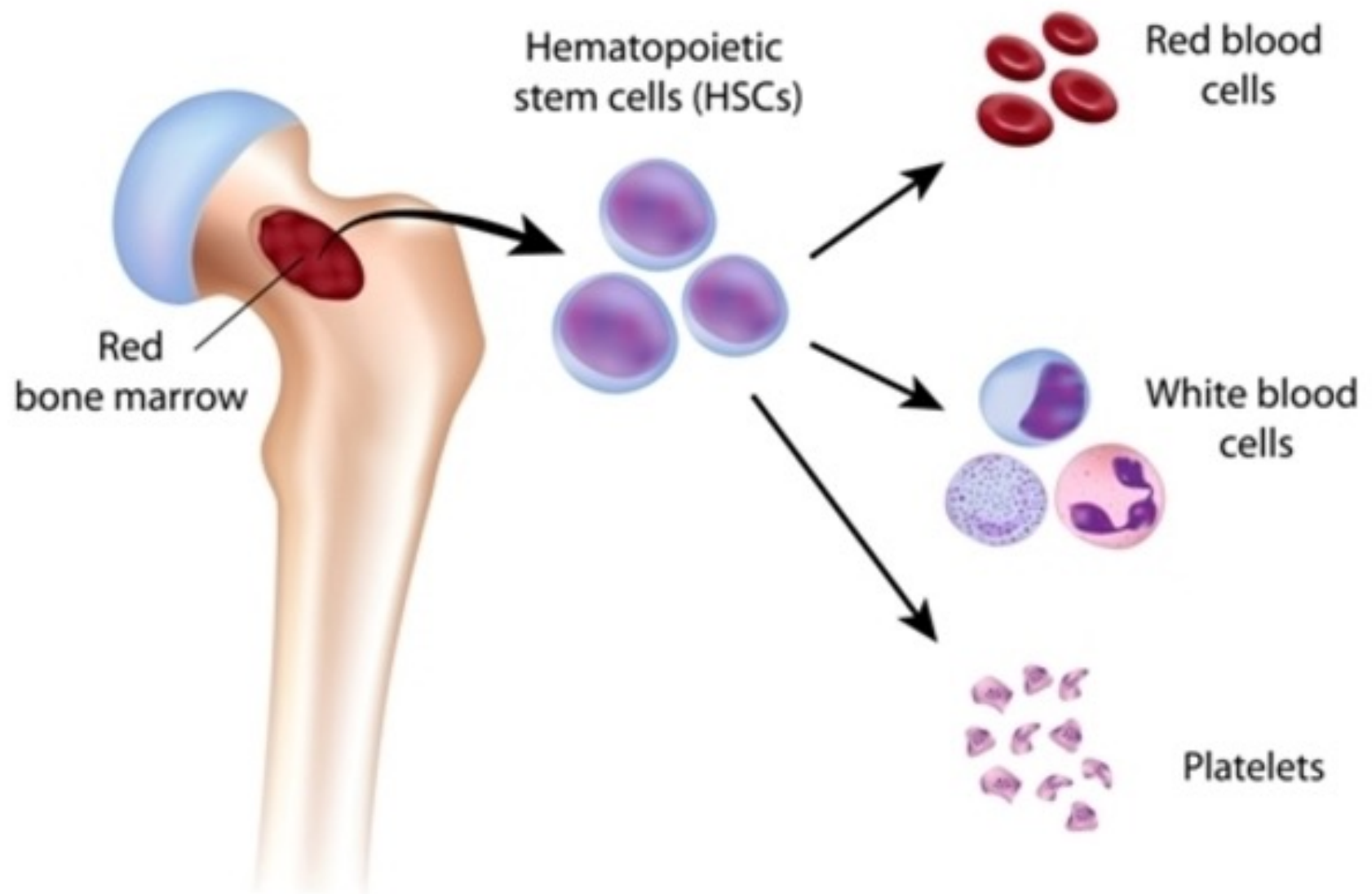
Red spots on skin

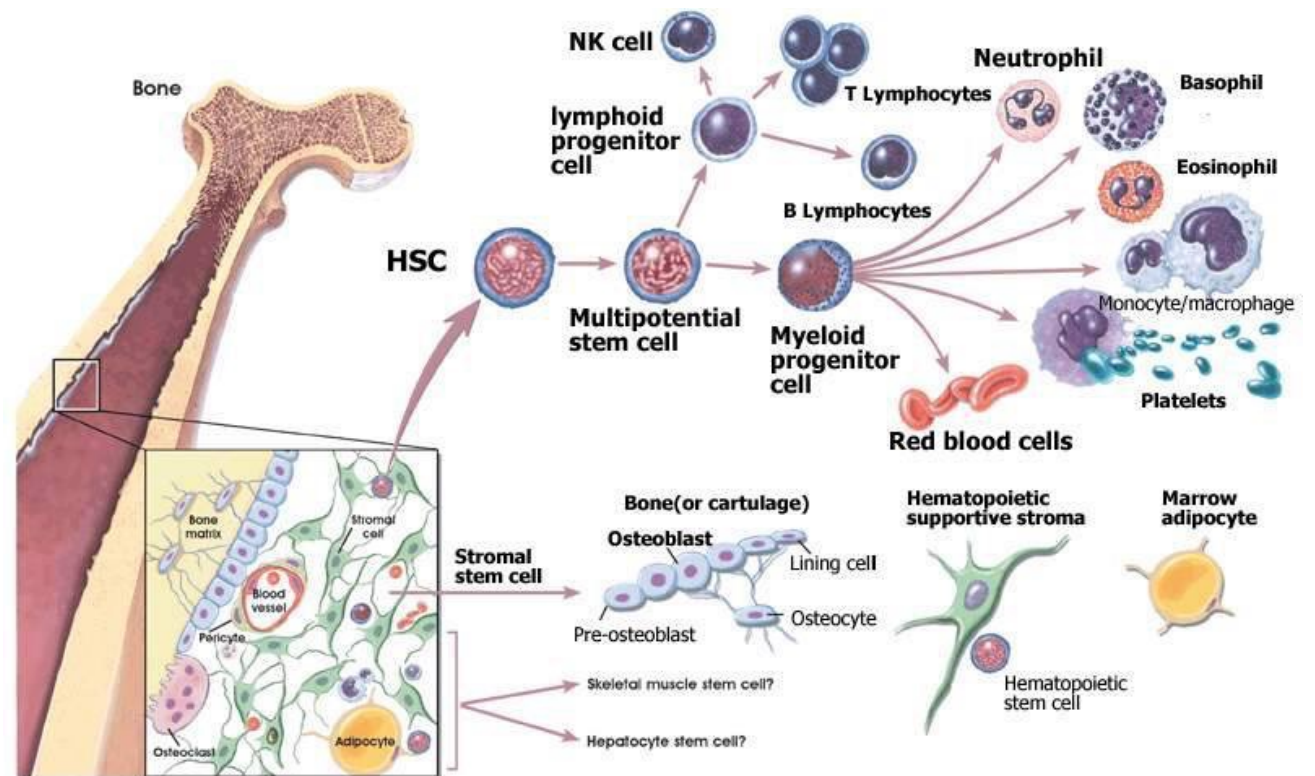
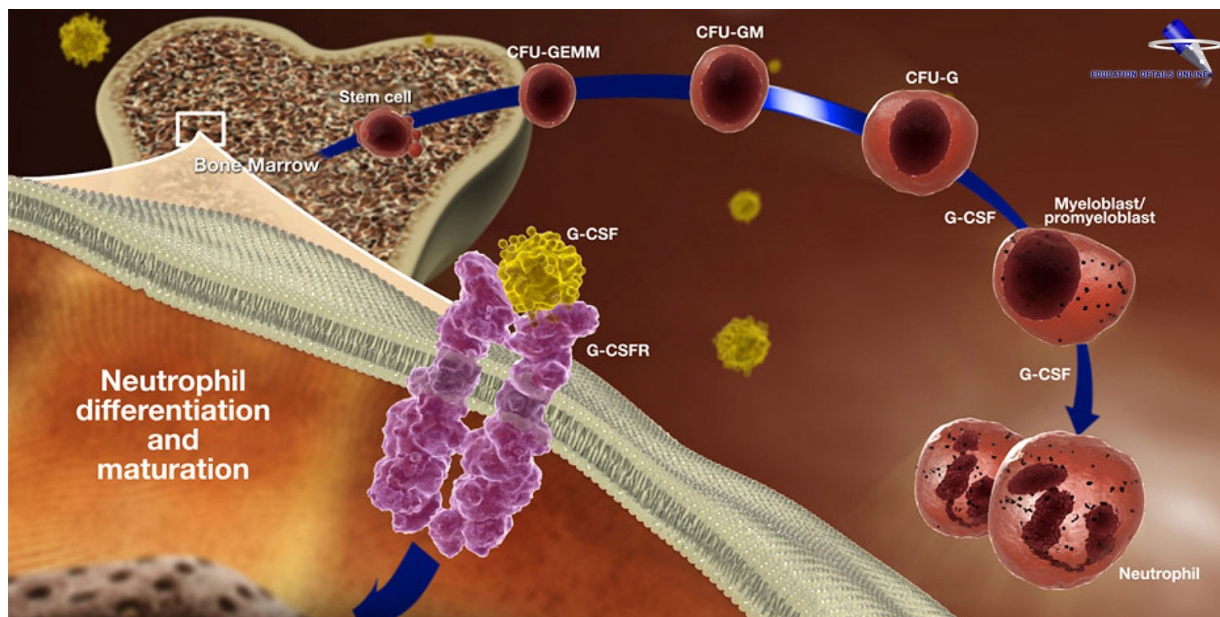


Spleen or liver Enlargement

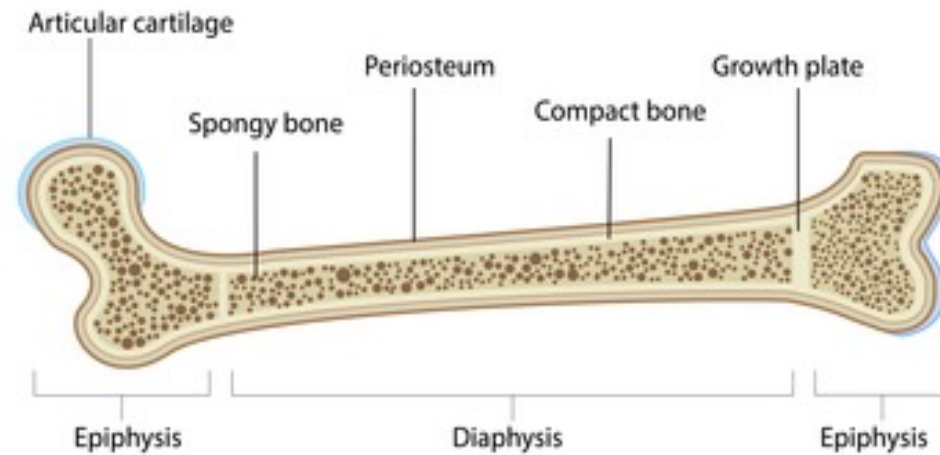






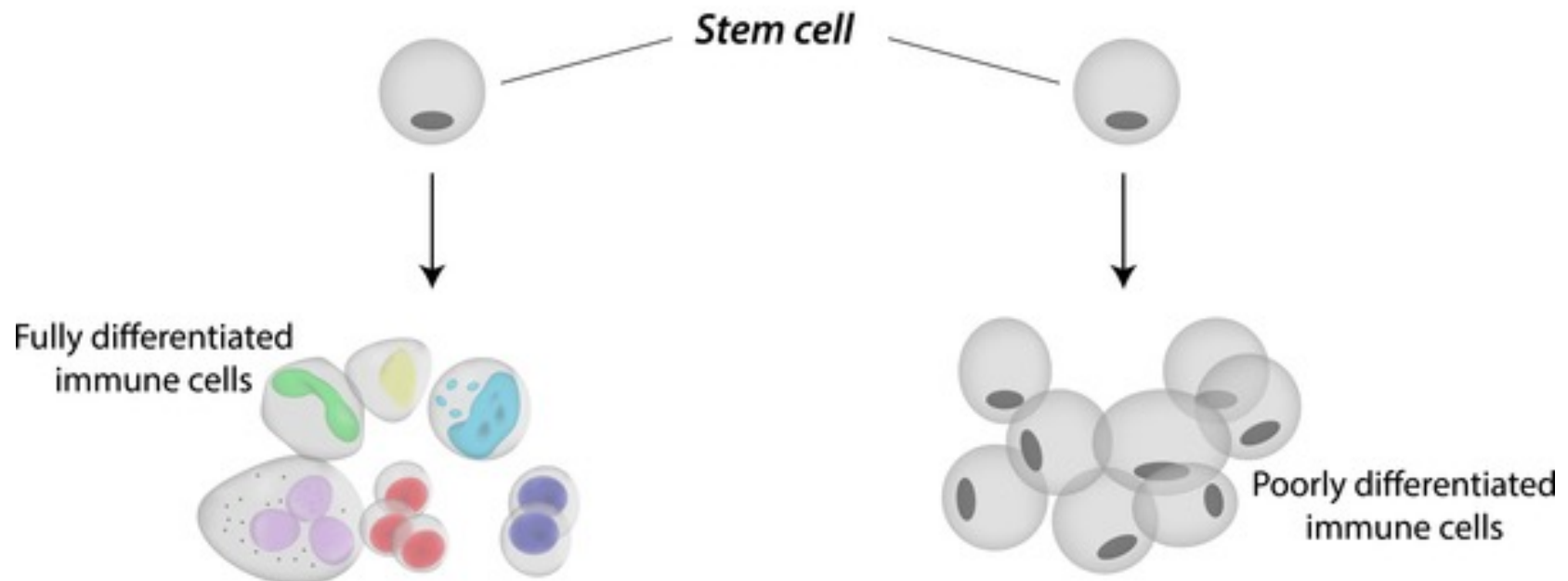


Leukemia



Healthy

Leukemia

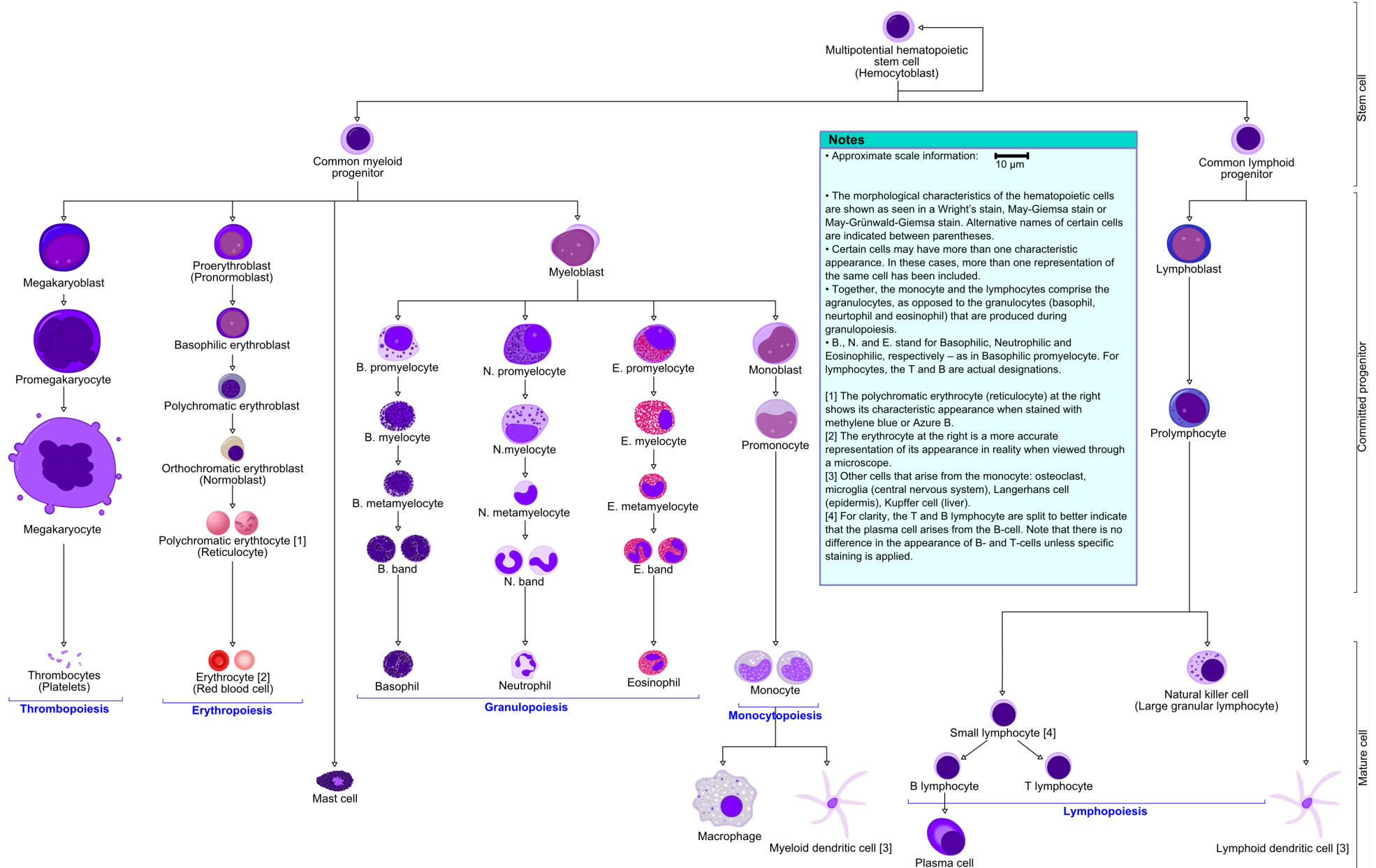


Hematopoiesis in humans

Bone marrow

Blood

Tissue



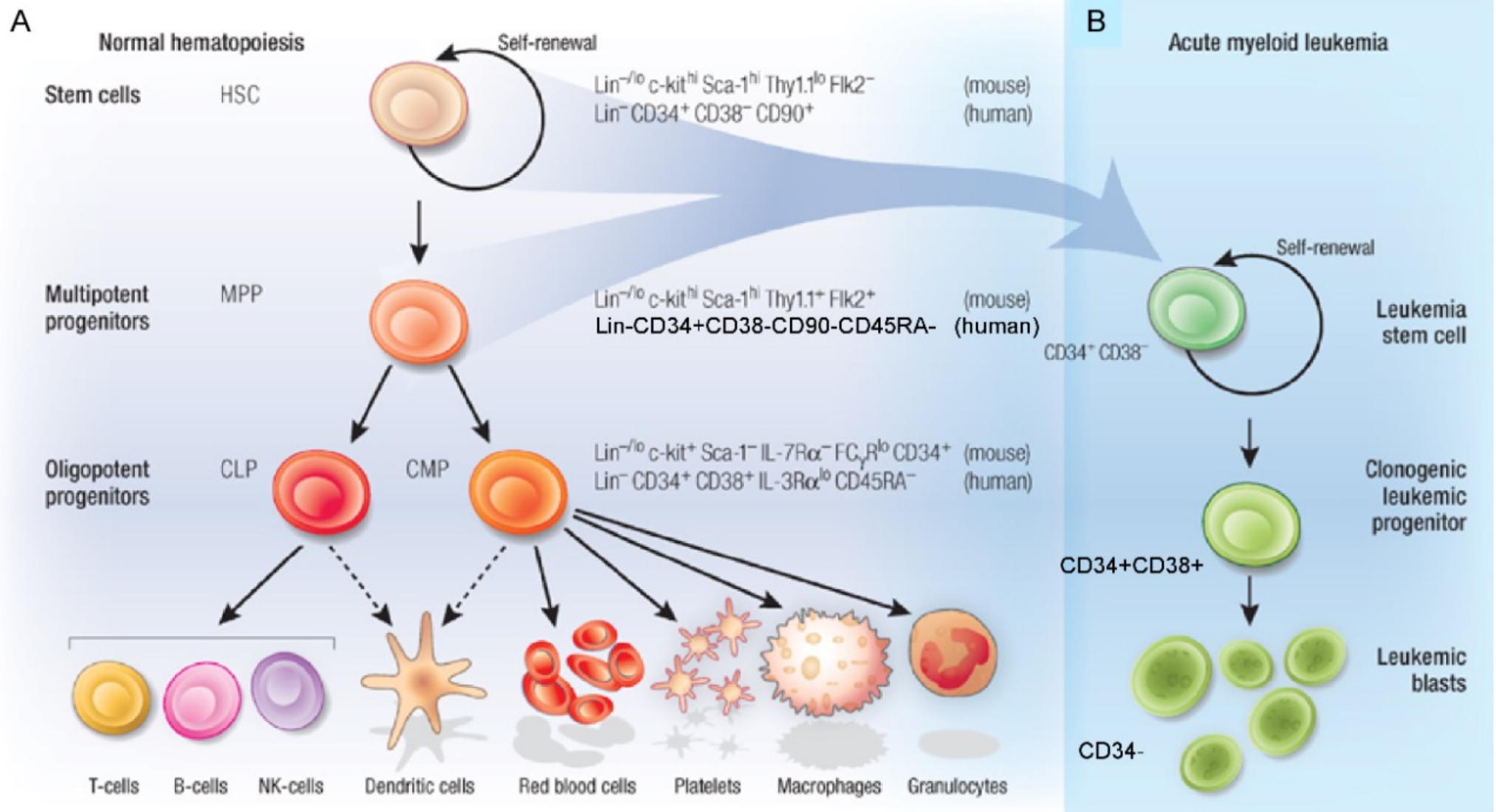


Figure : Hierarchical Model of Normal Hematopoiesis and Human Acute Myeloid Leukemia

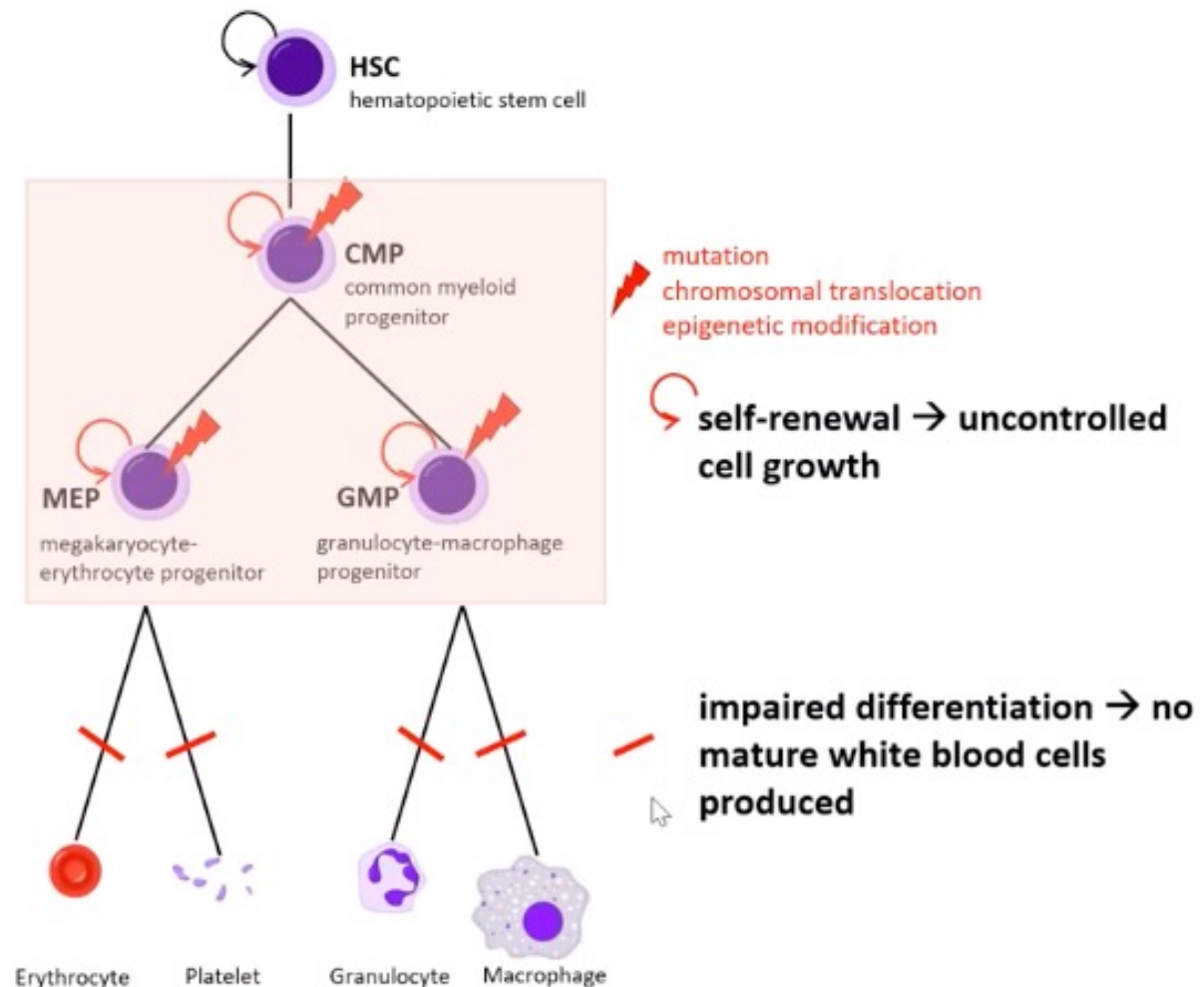
(A) Self-renewing HSC reside at the top of the hierarchy, giving rise to multipotent progenitors that in turn give rise to lineage-committed progenitors that eventually produce terminally differentiated blood cells. (B) AML is organized as a hierarchy initiated by self-renewing leukemia stem cells that give rise to leukemic progenitors, which in turn give rise to the leukemia blasts.

AML develops from immature white blood cells

Very early blood producing cells

Progenitor white blood cells

Mature blood cells

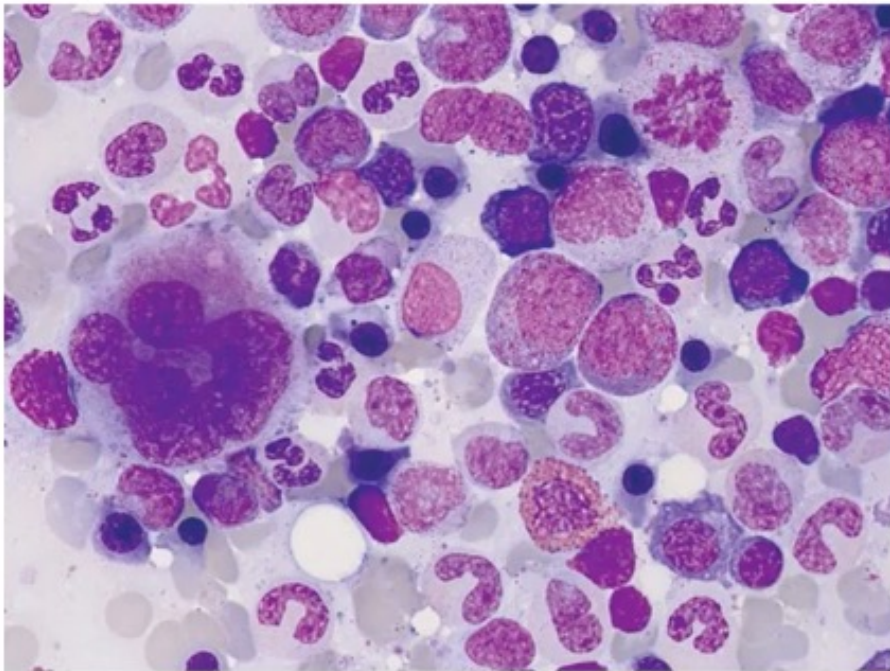


Phenotype for
disease

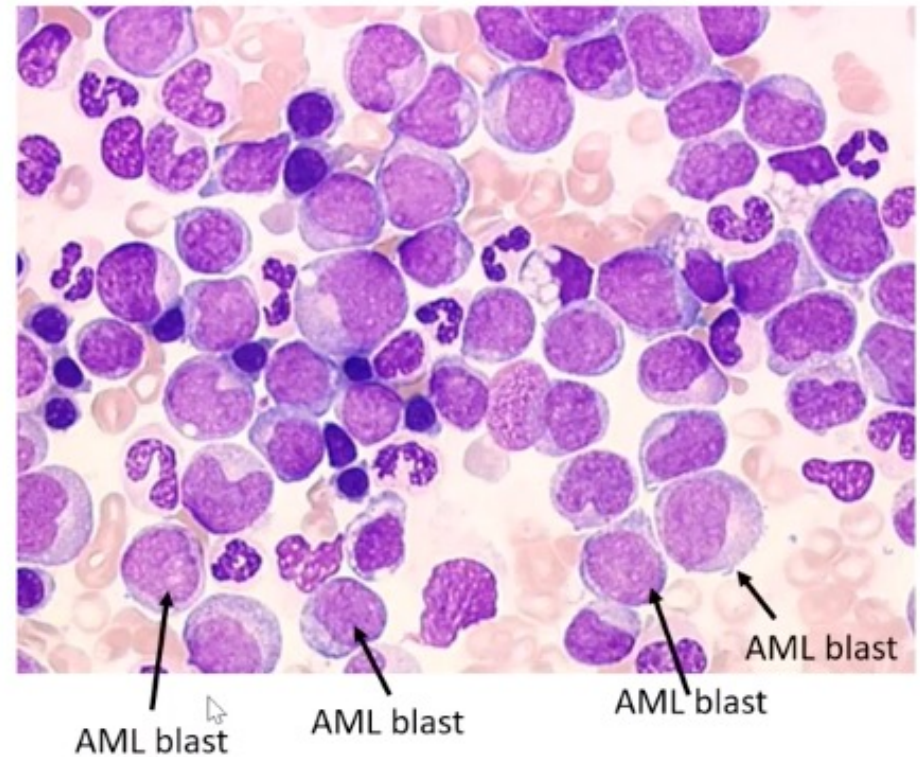


Normal BM vs AML

polymorph "colorful" picture



monomorph picture

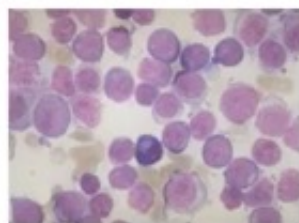


≥20% blasts in the bone marrow (BM) or peripheral blood

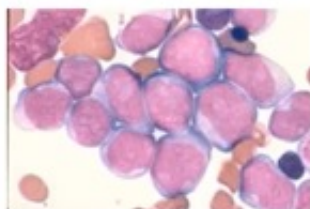
Classification of AML: FAB

Based on Morphology: **F**rench**A**merican**B**ritish classification (1976)

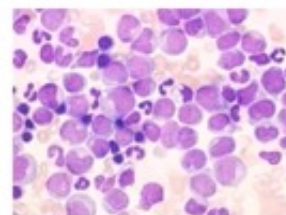
M0



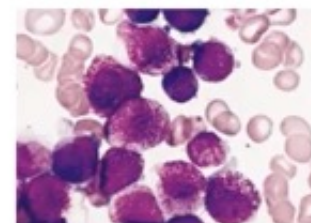
M1



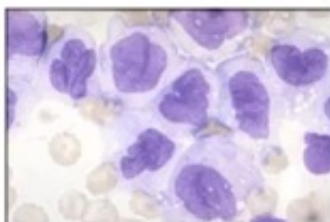
M2



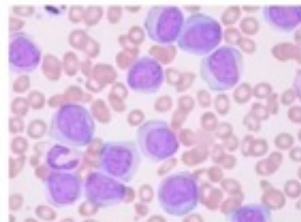
M3



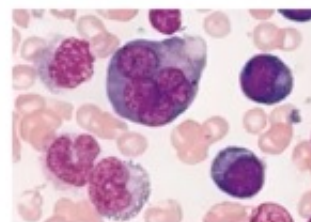
M4



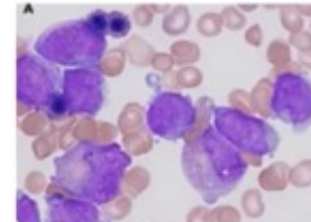
M5



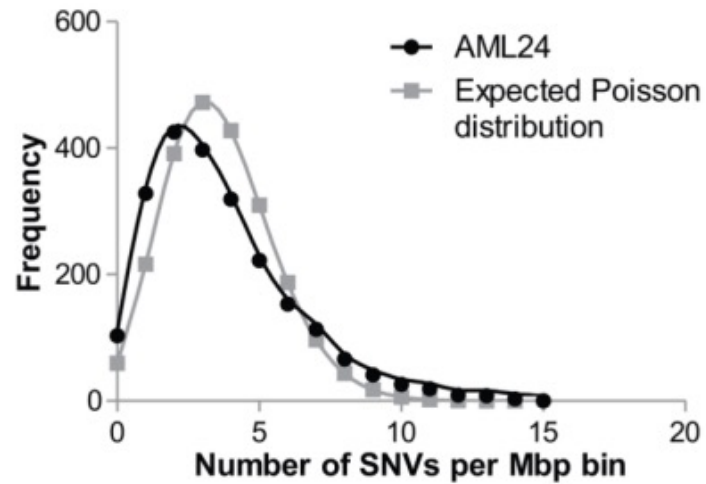
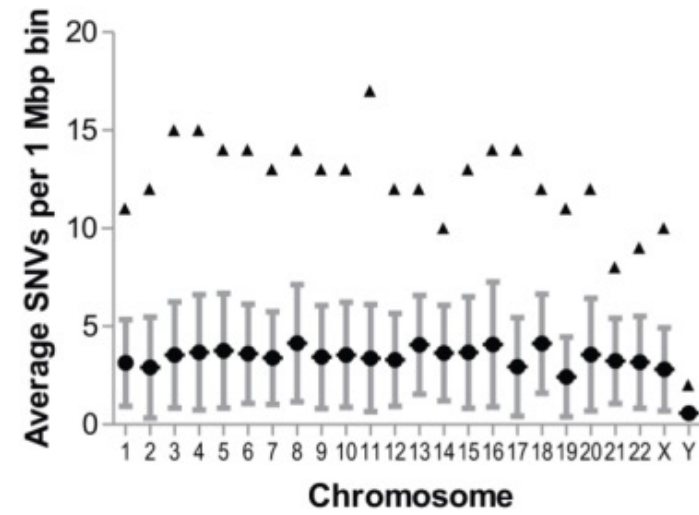
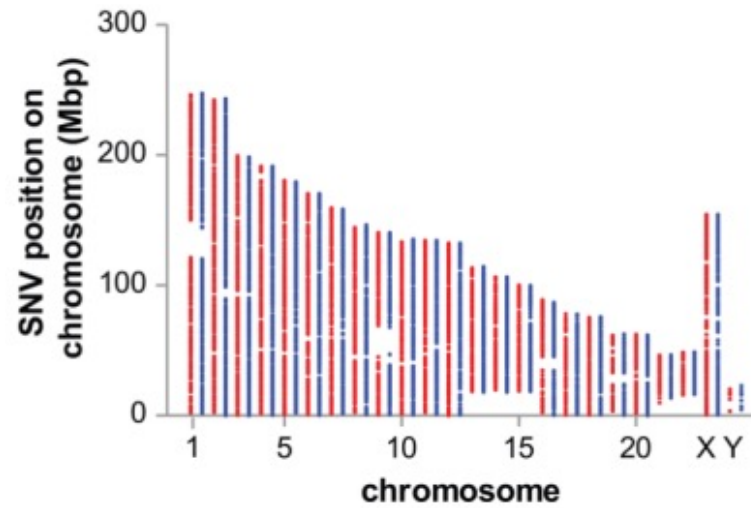
M6

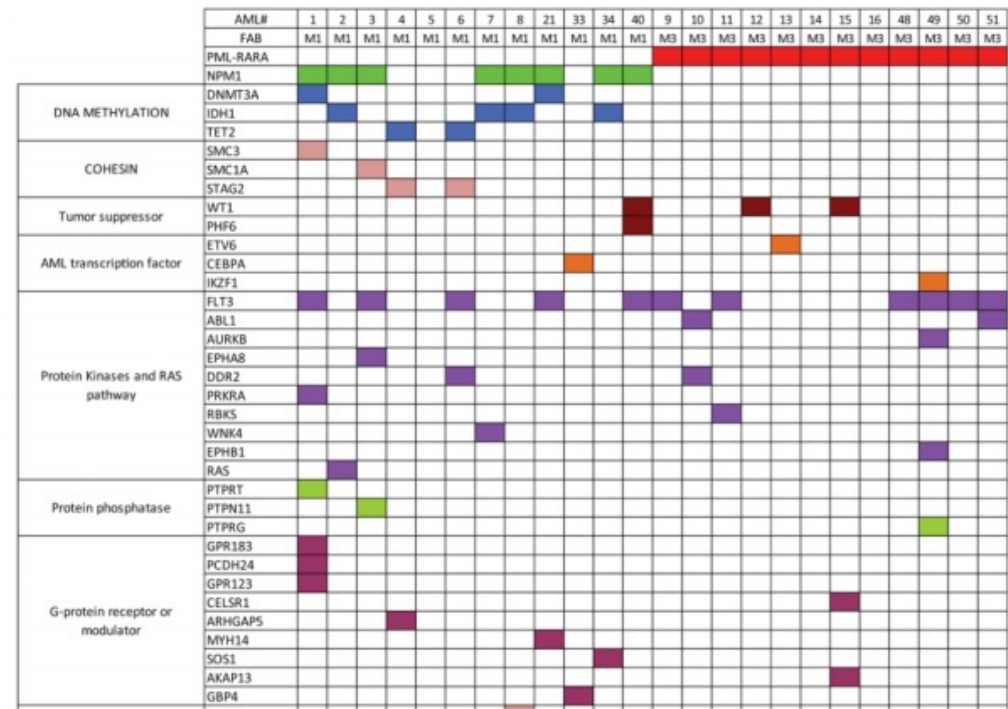
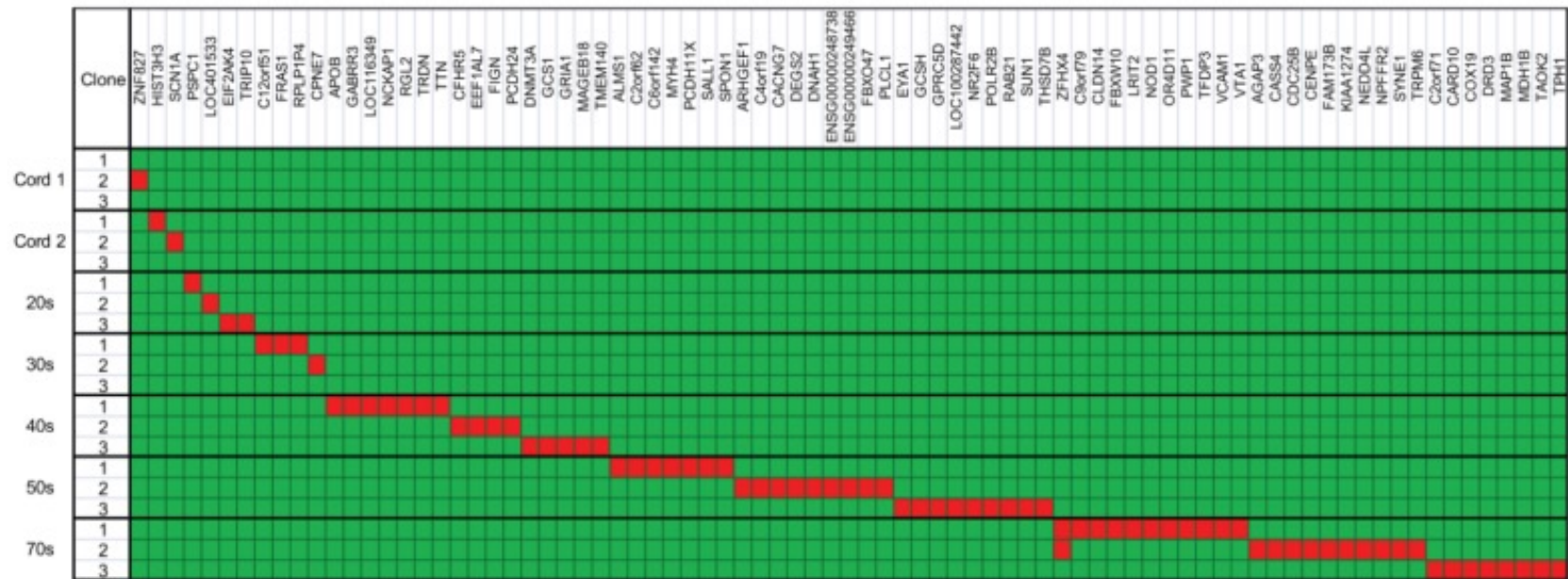


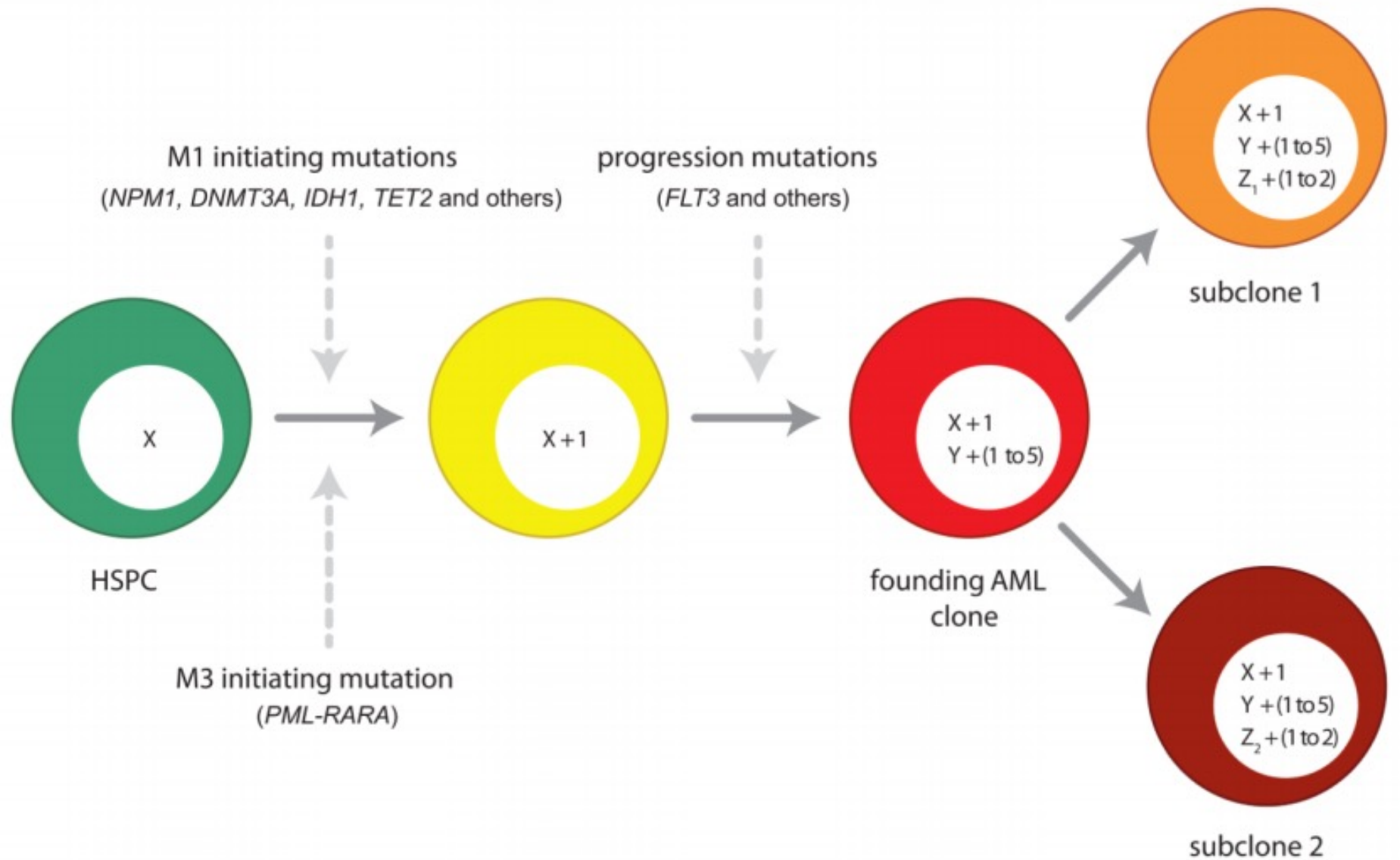
M7



Genome sequencing

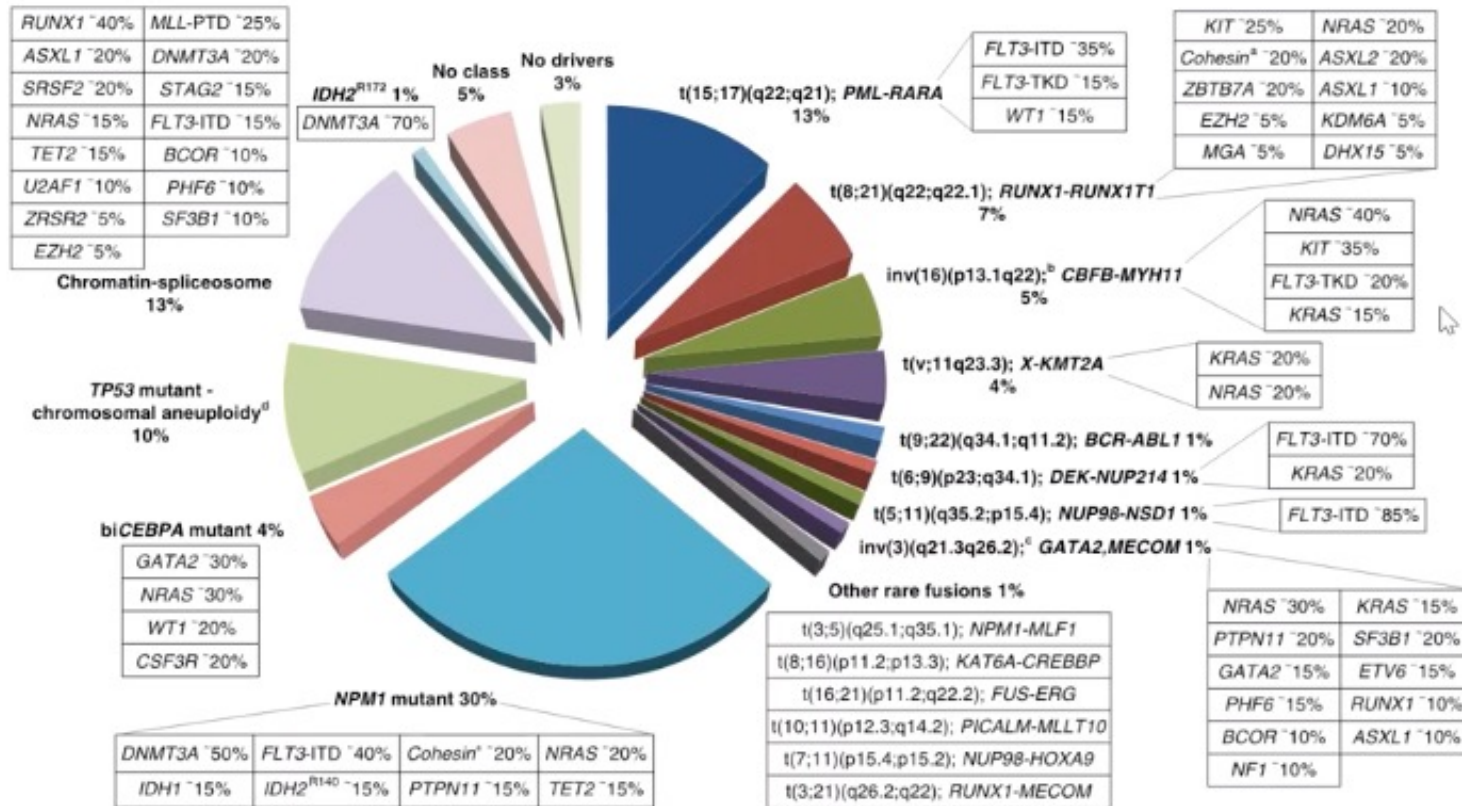






X : passenger mutations pre-existing in HSPC
 Y : passenger mutations gained between initiating and progression events
 Z : passenger mutations gained during progression to subclones

AML is a genetically heterogeneous disease

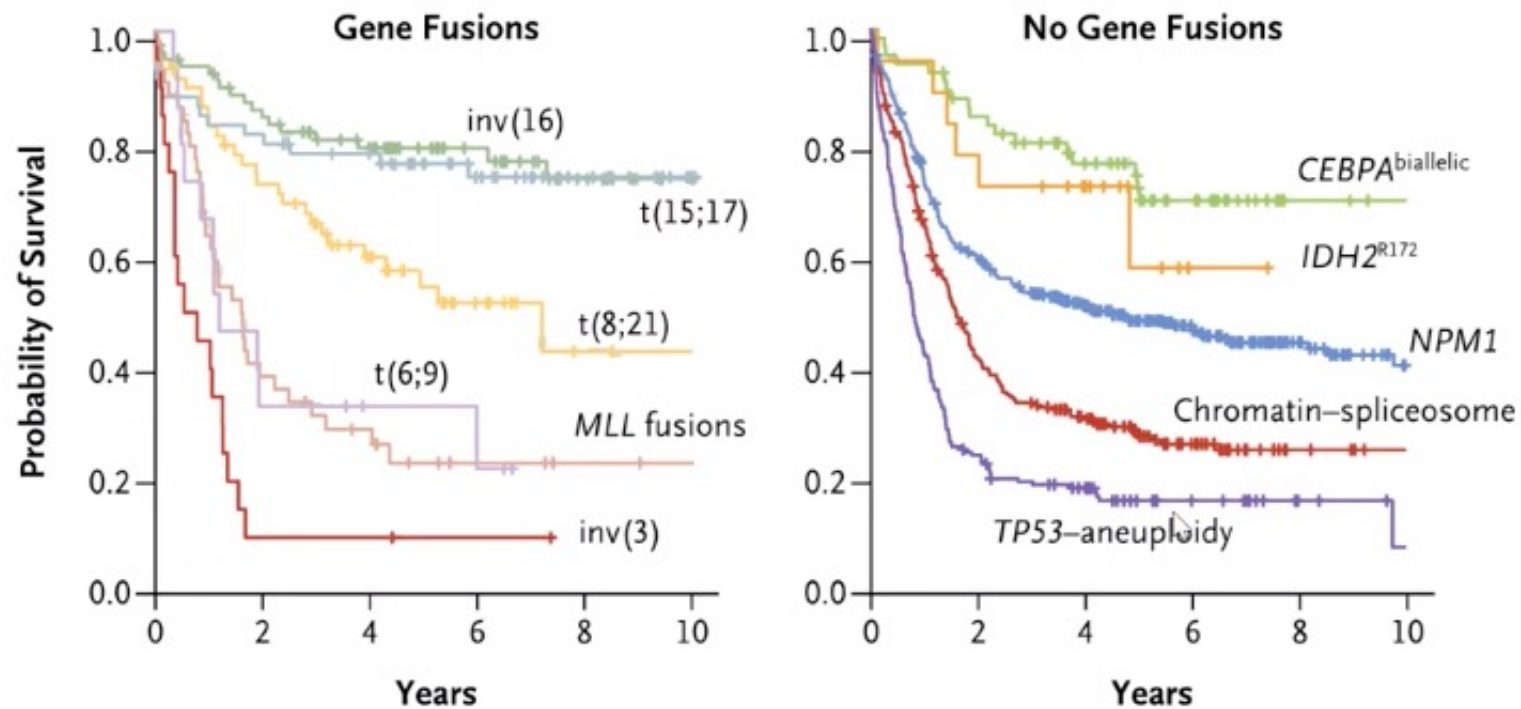




Diagnostic based on Genome variations

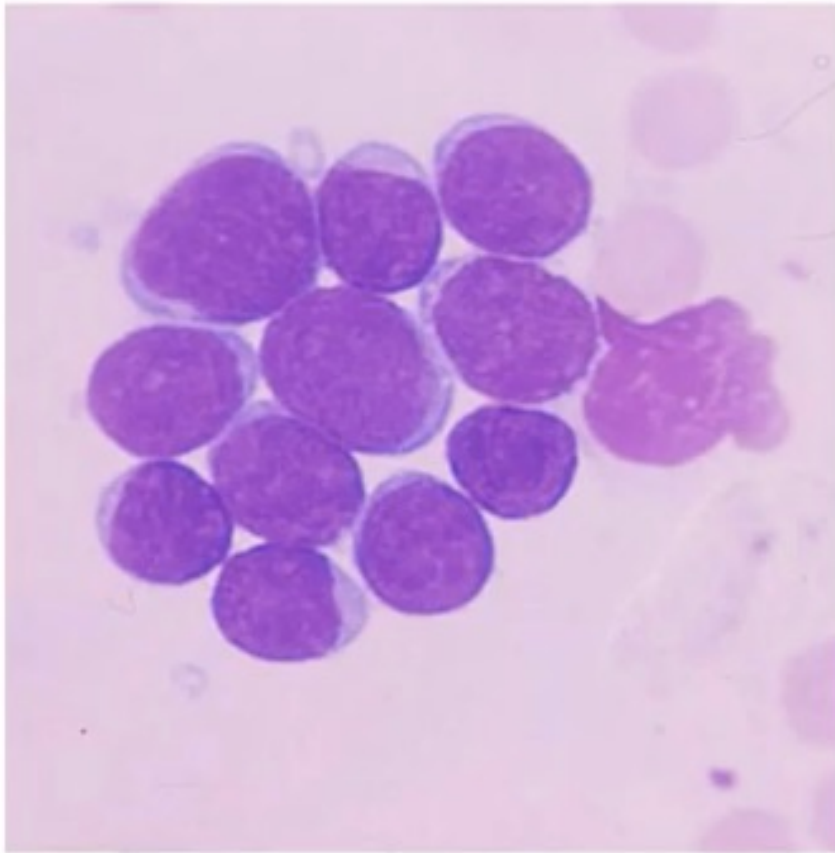


Genetics determine overall survival

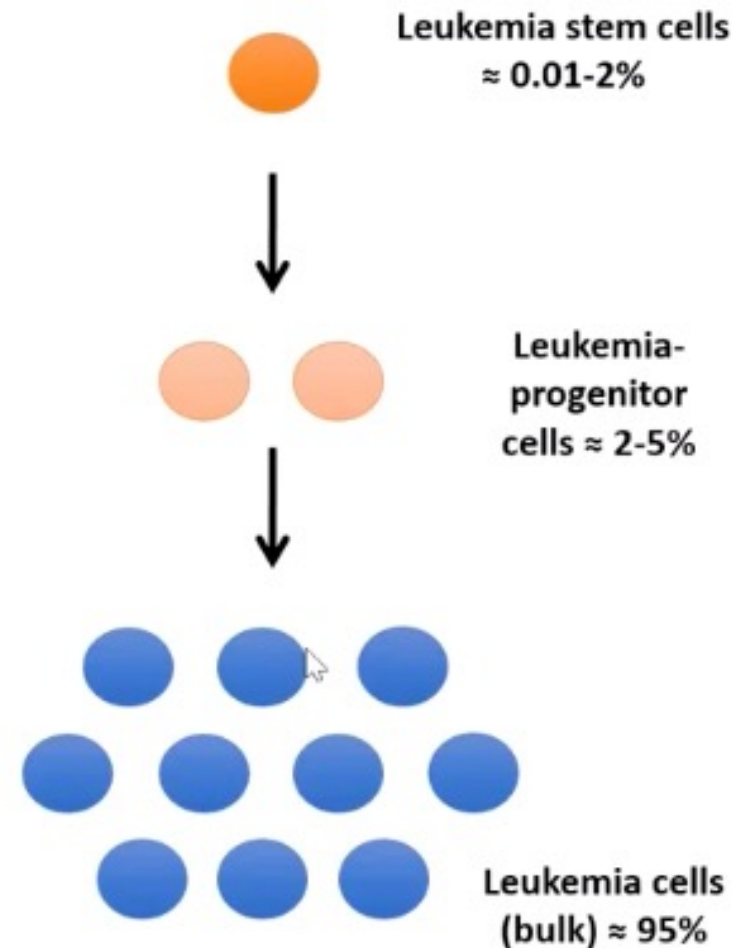


AML is a stem cell disease

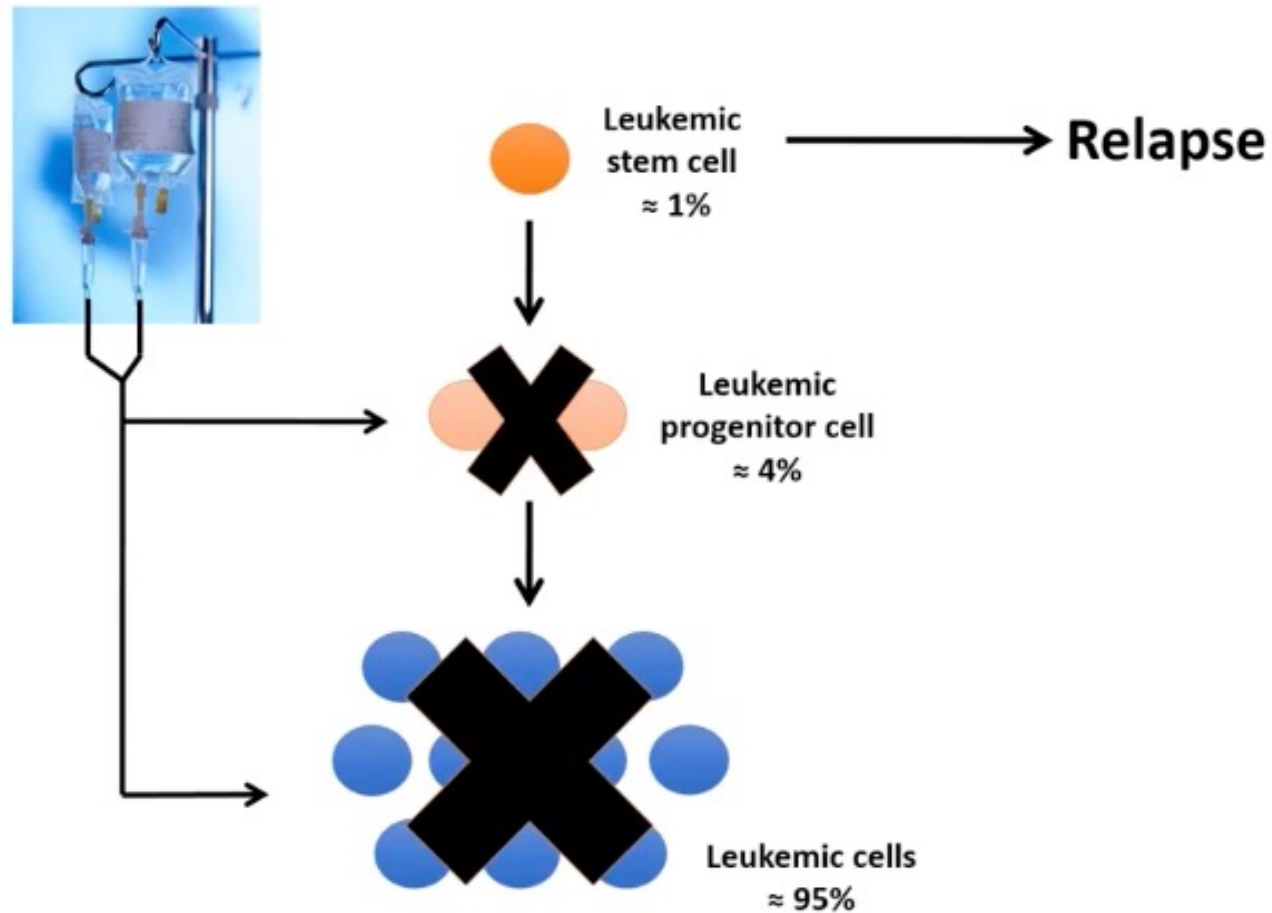
AML cells look similar



...but are not

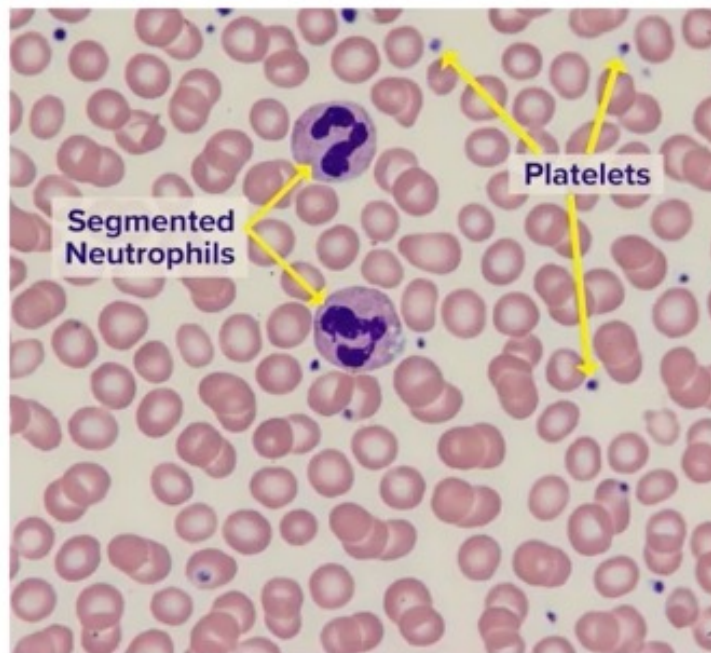


Leukemic stem cells impact on treatment success

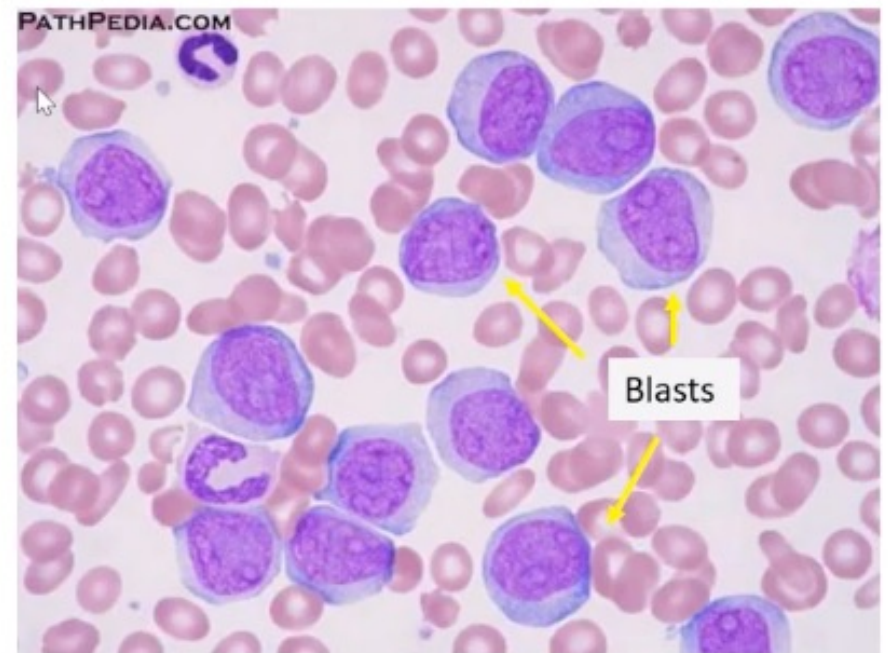


Differential blood count/Blood smear

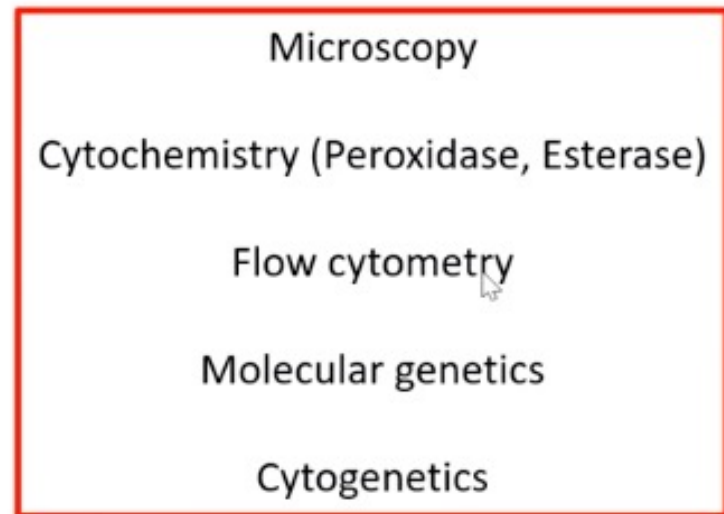
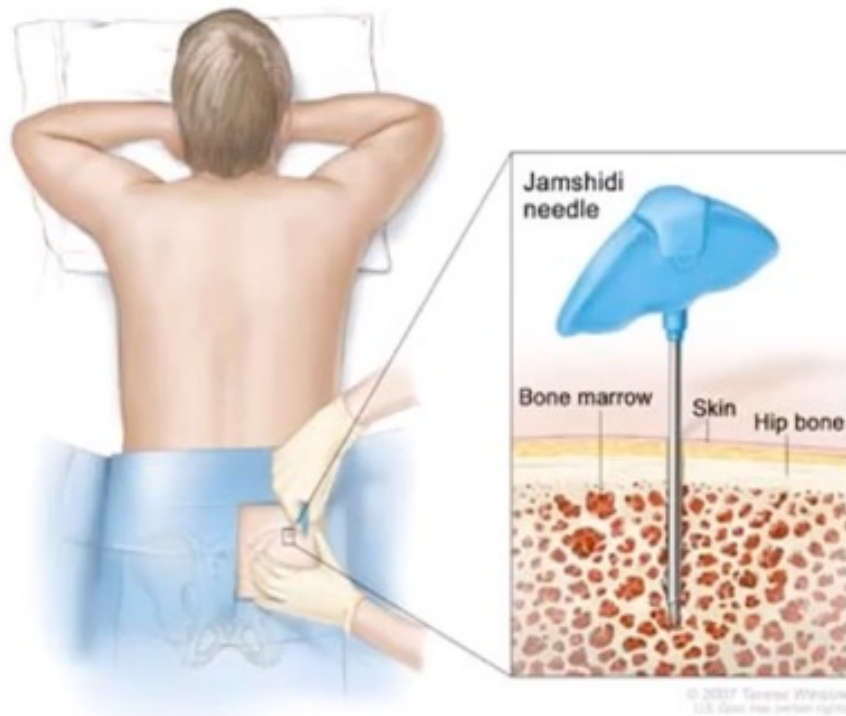
Healthy



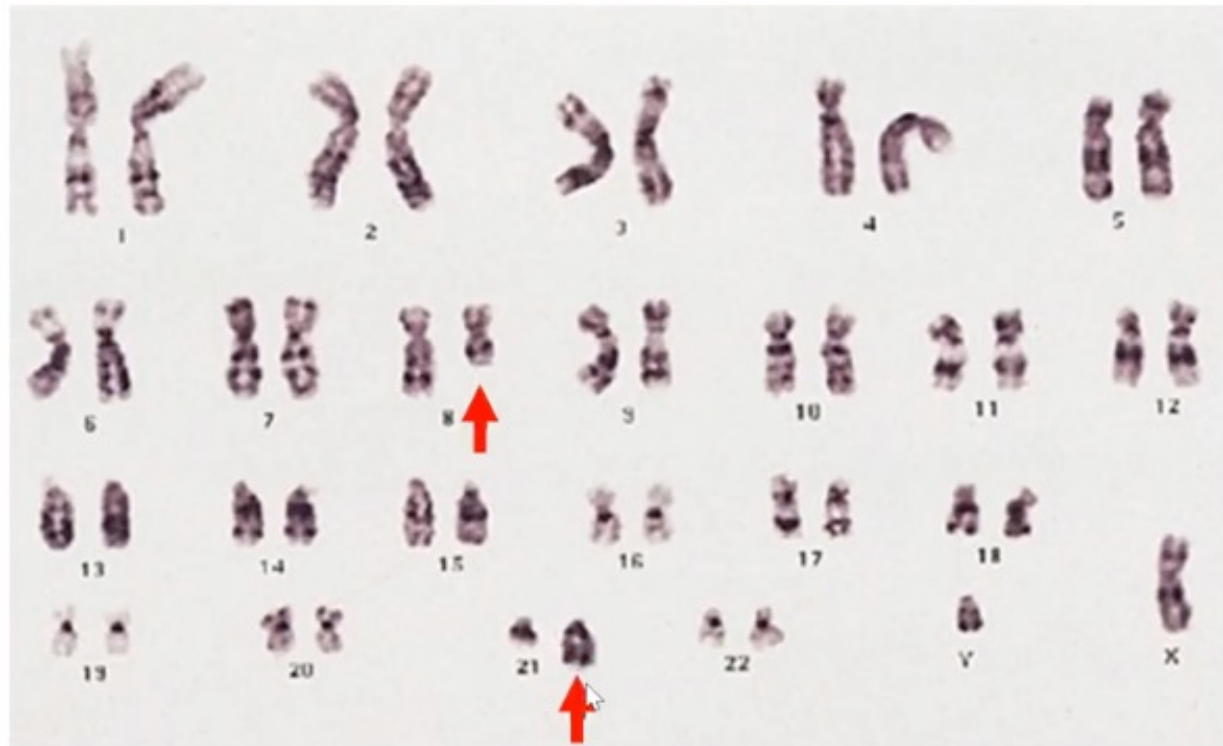
AML



Bone marrow aspiration and biopsy



Cytogenetics



t(8;21)(q22;q22)

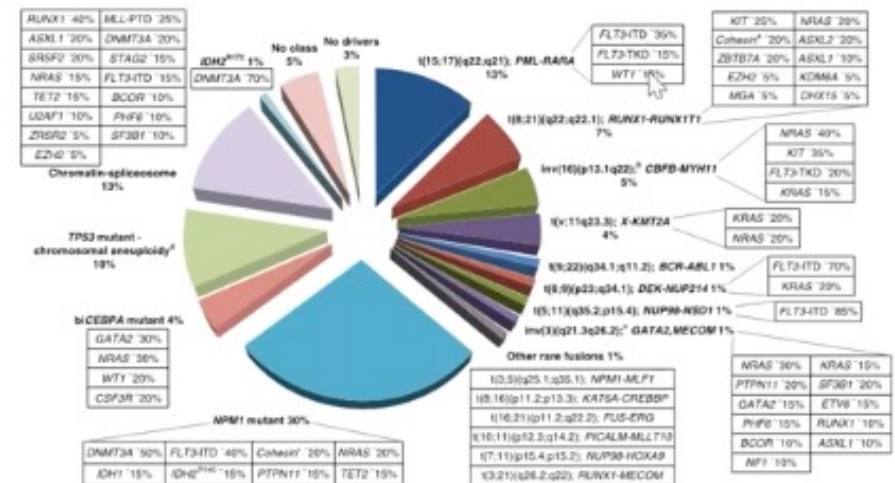
AML1/ETO

Molecular genetics



DNA Sequencing

+ Cytogenetics =



AML subtype

Features of APL

Characterized by

translocation **t(15;17)**

Very good prognosis

Treatment “almost” chemo-free

Risk groups

low (WBC ≤ 10, plts > 40),

intermediate (WBC ≤ 10, plts ≤ 40),

high (WBC > 10)

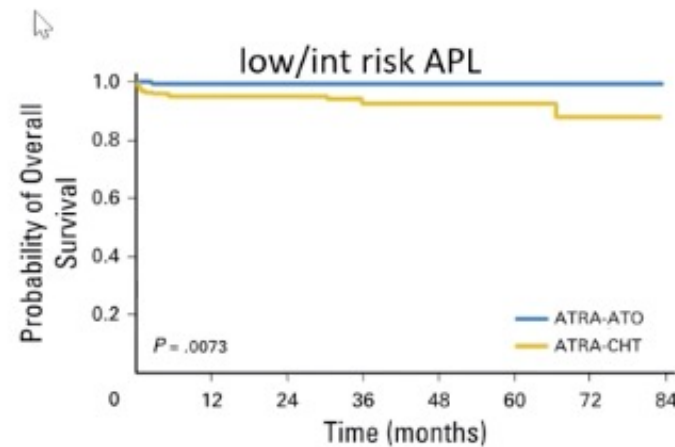
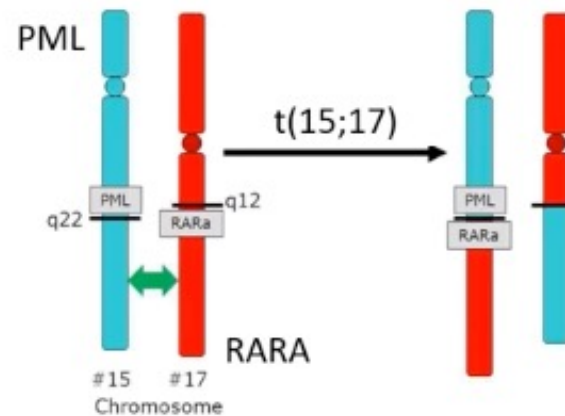
Treatment

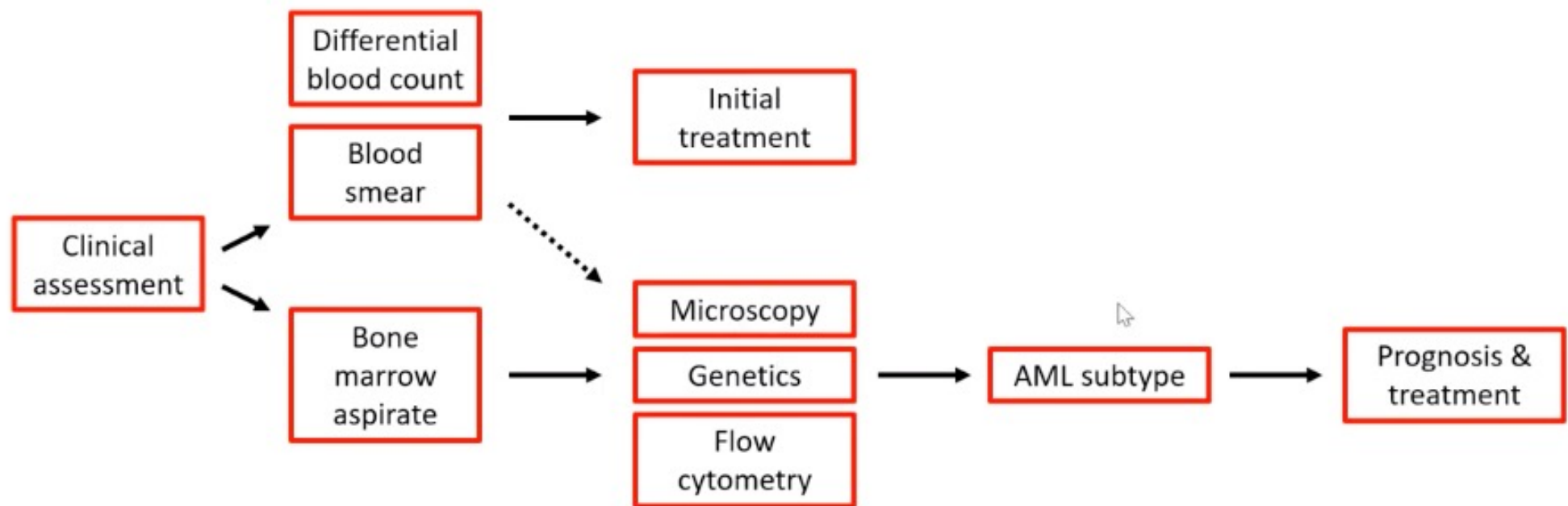
ATRA+ATO ± Chemo

Problems

Differentiation syndrome

Coagulopathy

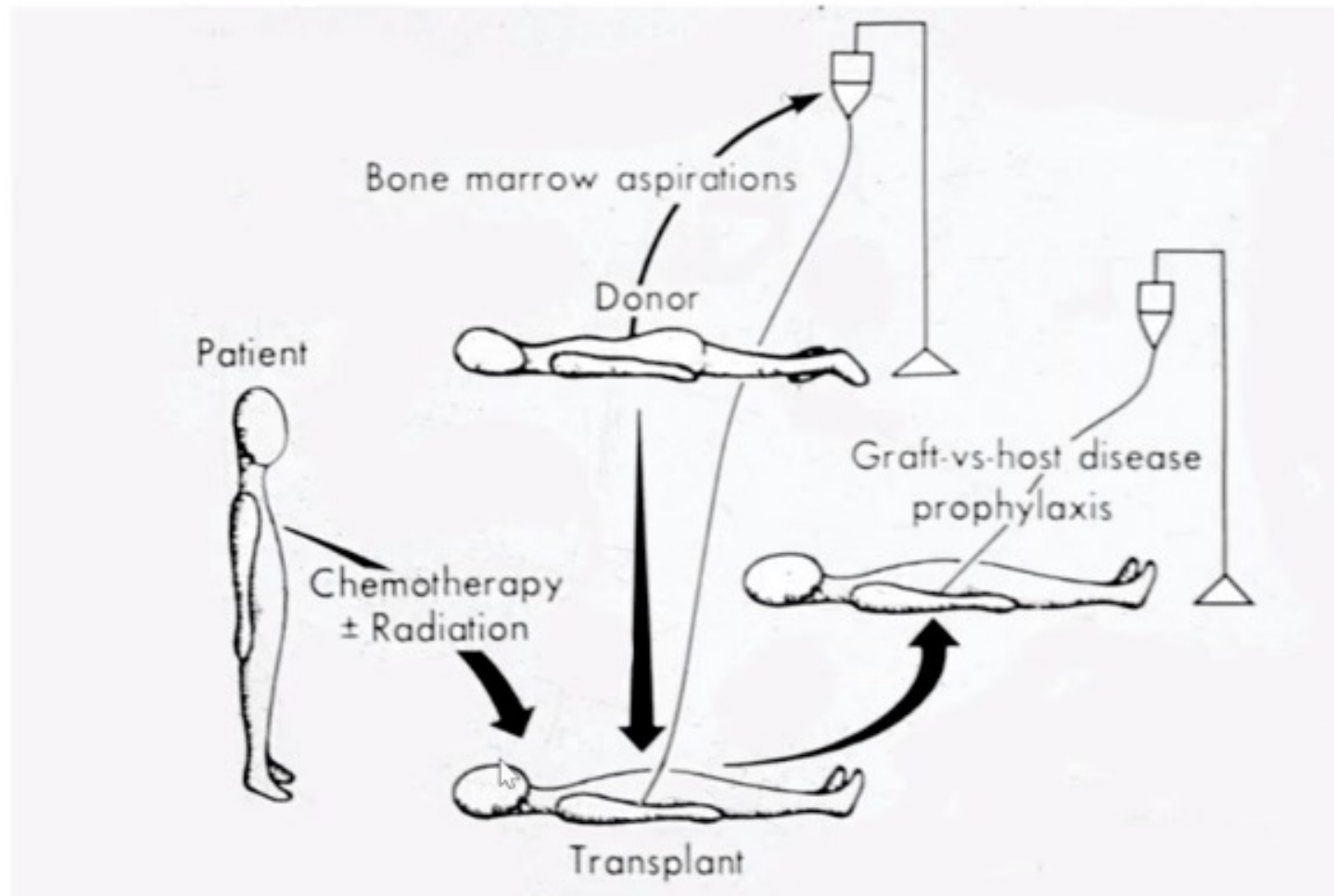






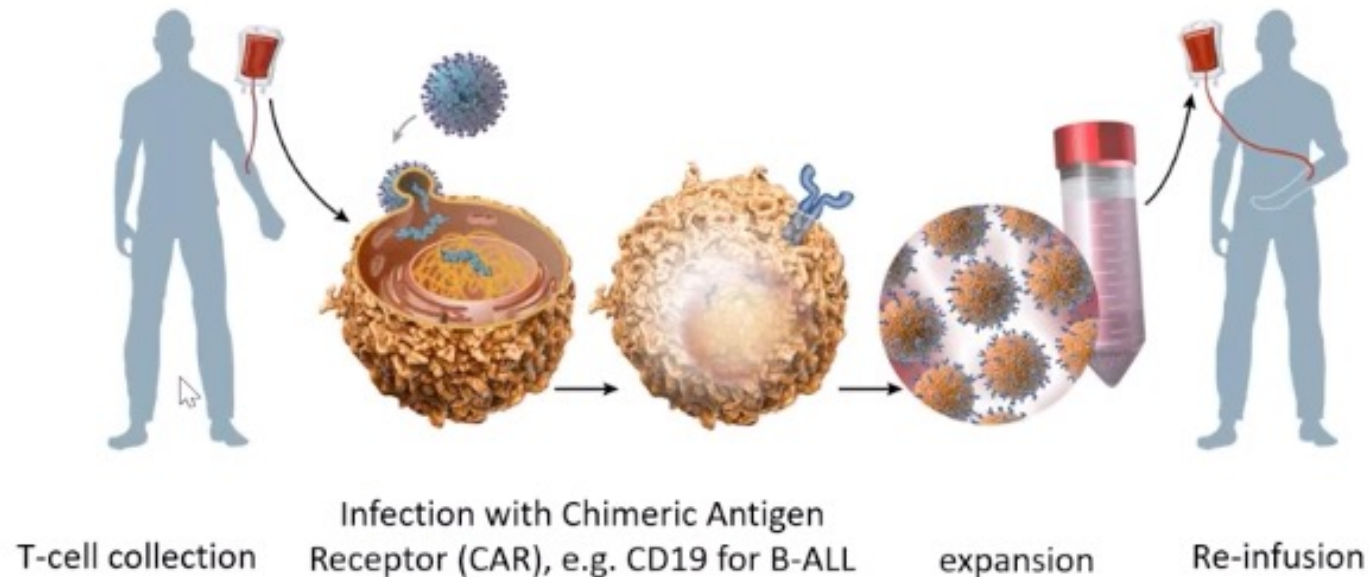
Treatment strategies

Allogeneic stem cell transplantation



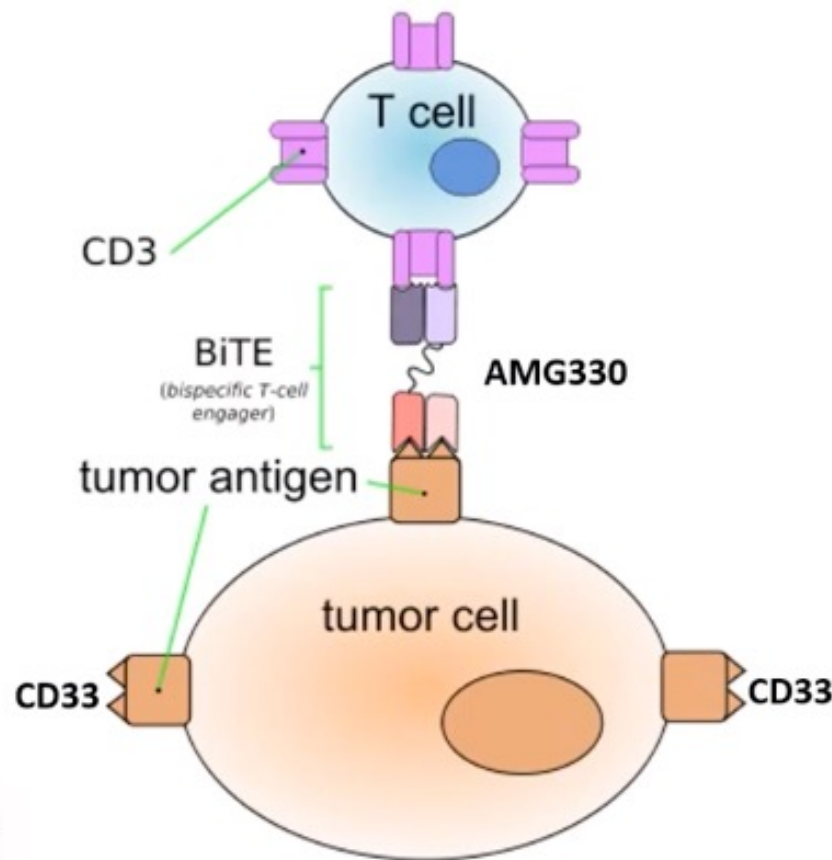
- Mother of all immunotherapies
- First allo HCT 1957
- 80-100 allo HCT in British Columbia
- Age limit in BC $\leq 70y$

How CART cell therapy works



- Tailored therapy
- Highly effective independent of cytogenetics, penetrate CSF
- Side effects: cytokine release syndrome, neurological side effects
- Expensive and limited availability

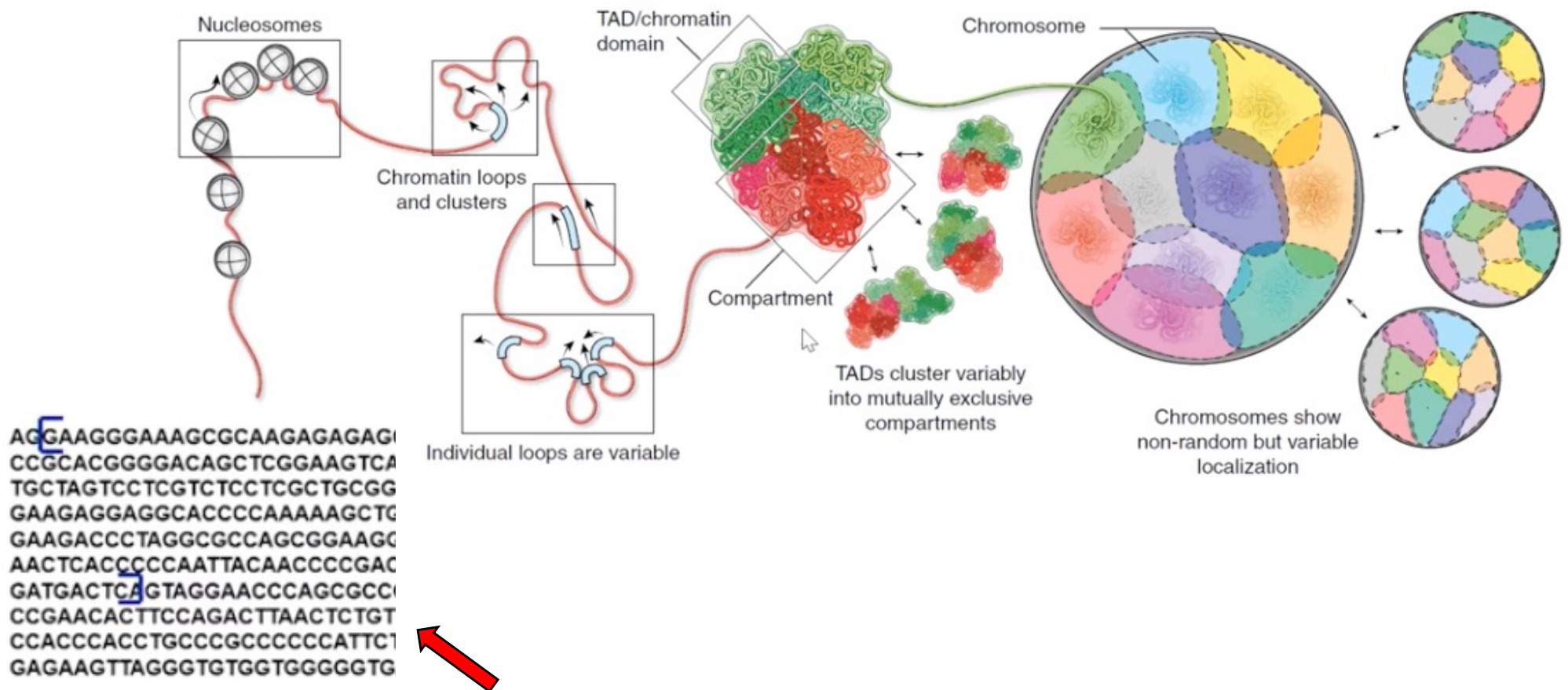
Other immunotherapy approaches

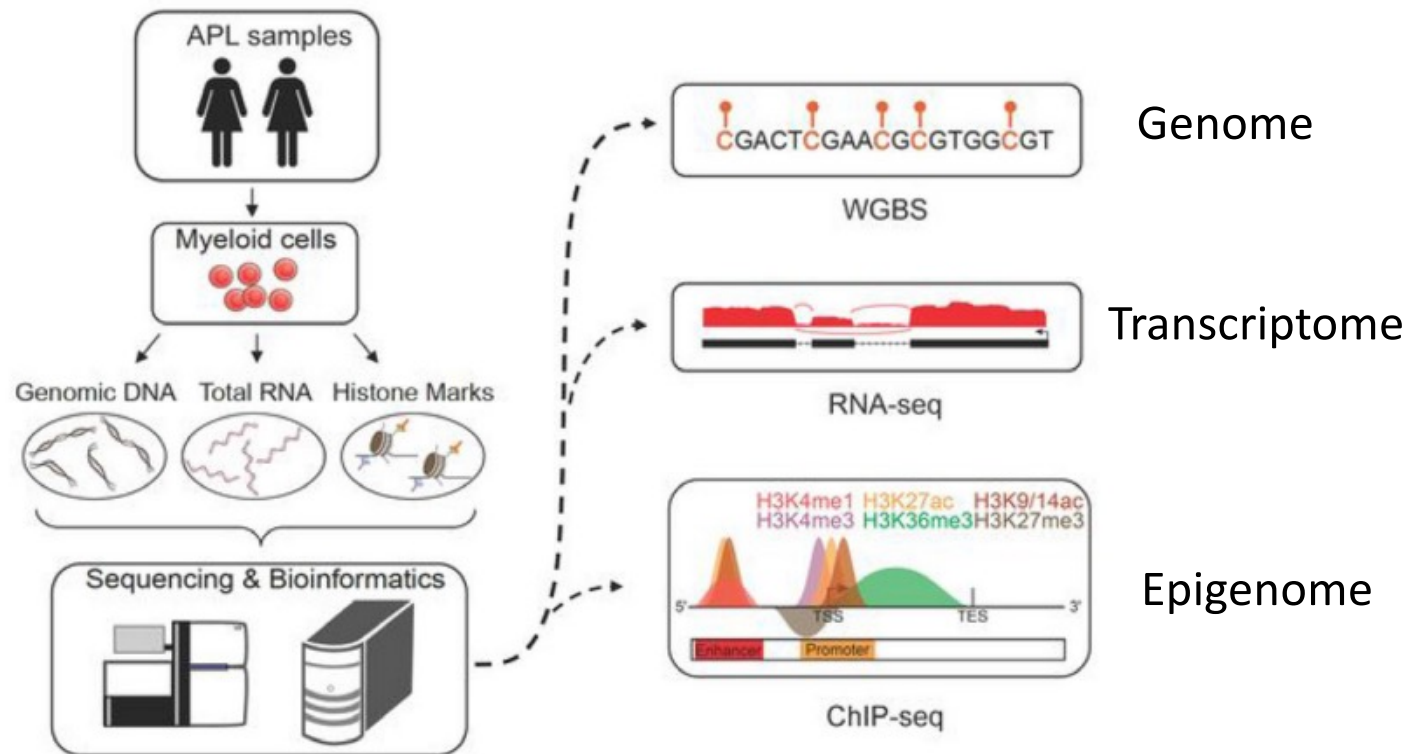


- Bispecific antibody AMG330
- First dose finding trial (Phase I)
- Highly pretreated patients
- Almost all patients eventually progressed
- Complete response in 2 patients

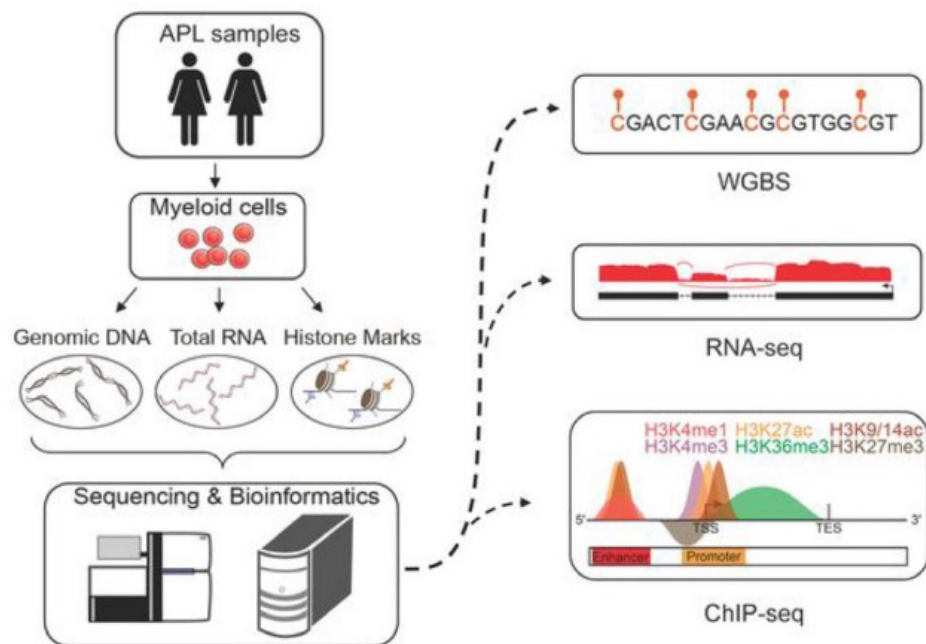


Multi-omics

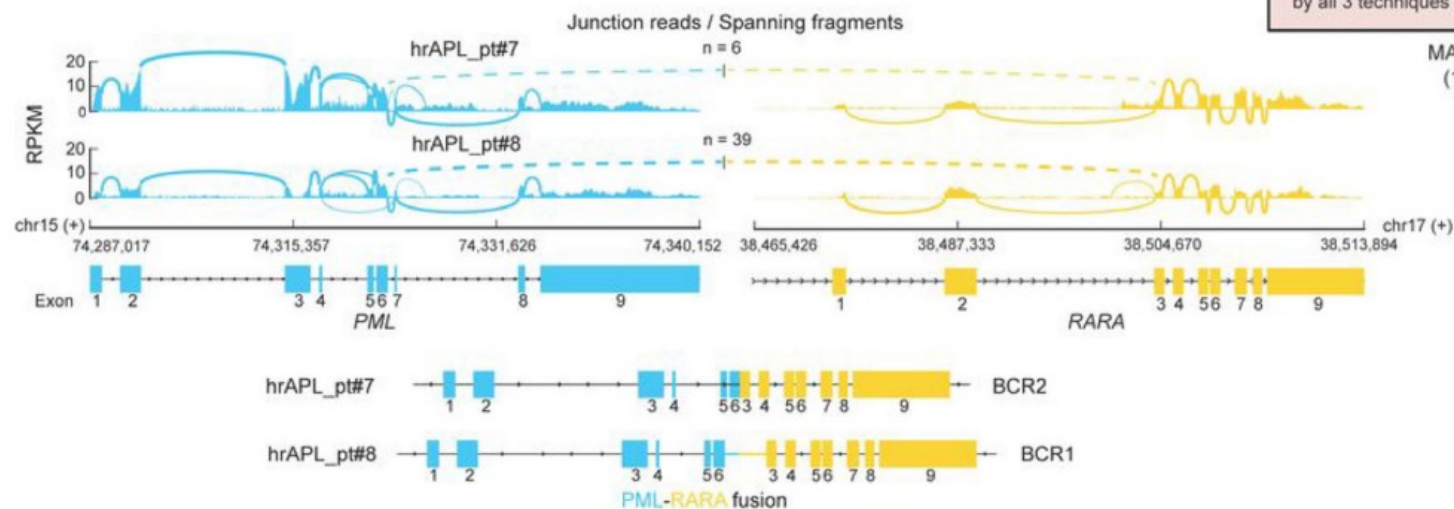




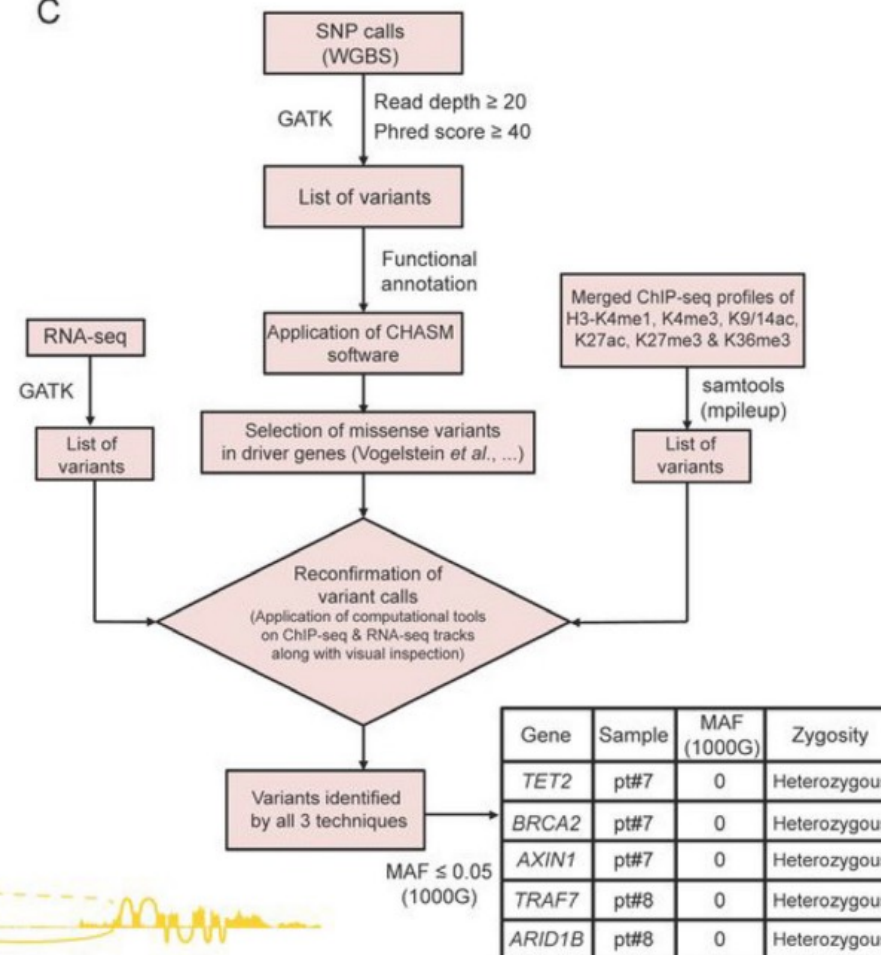
A

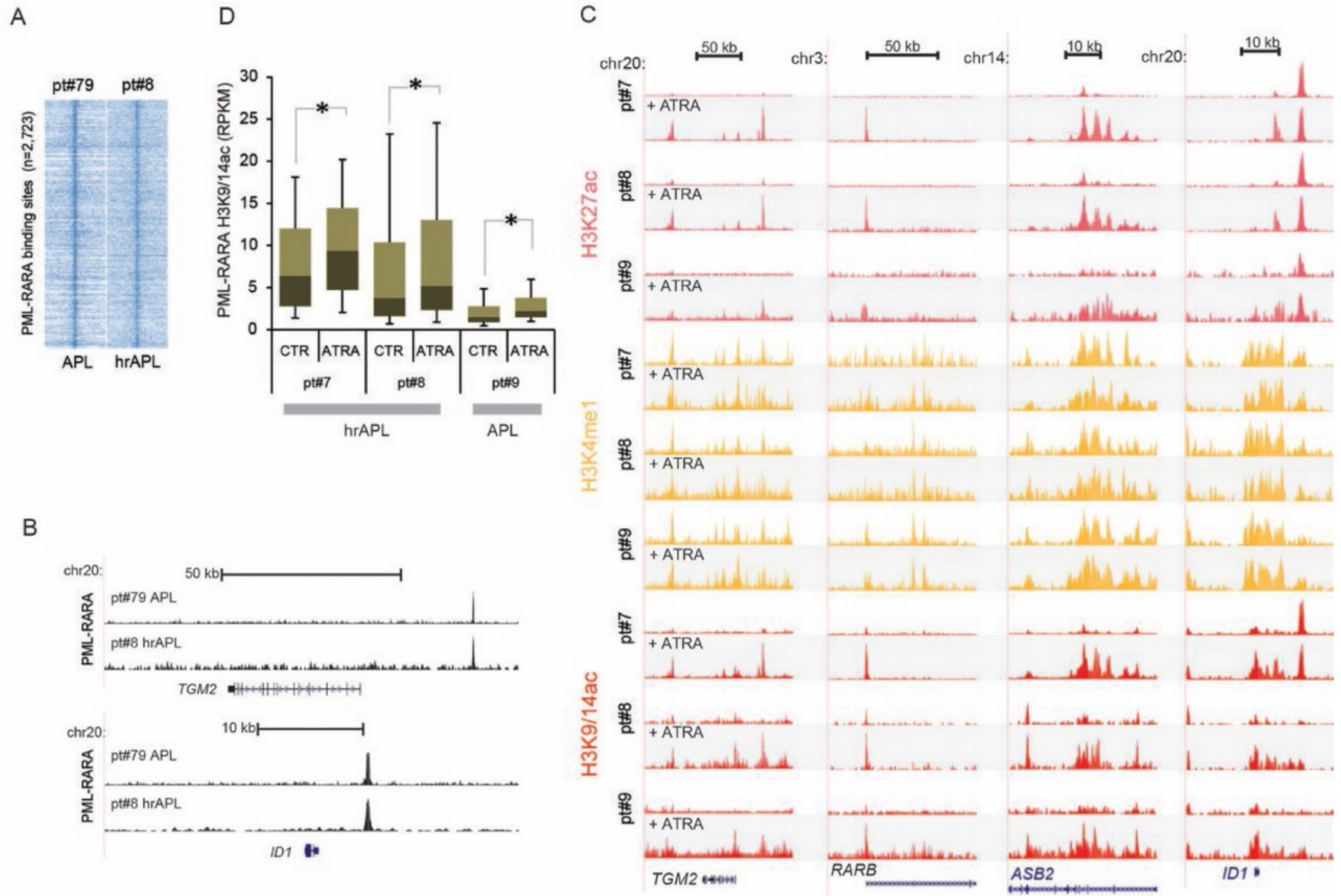


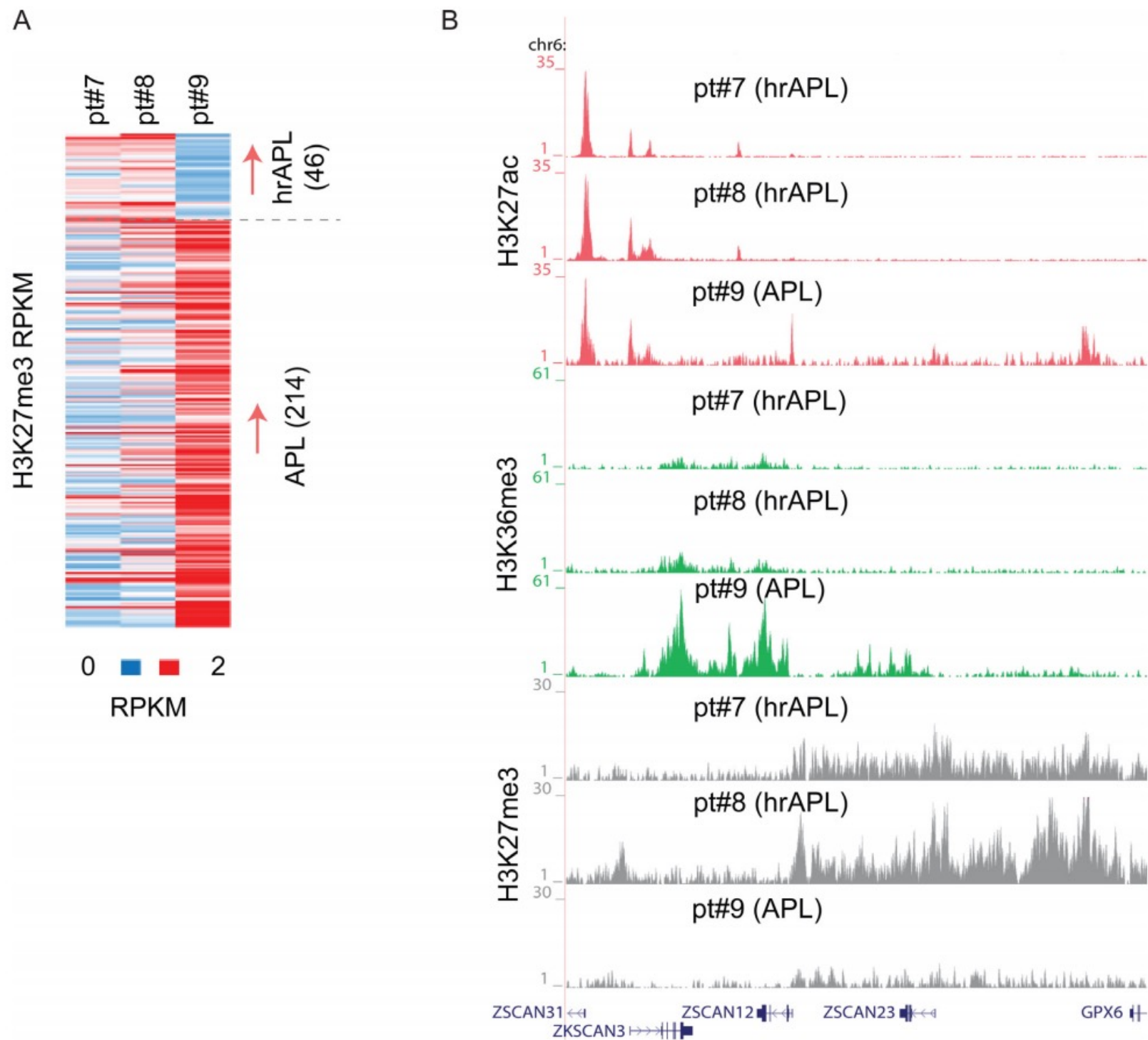
B

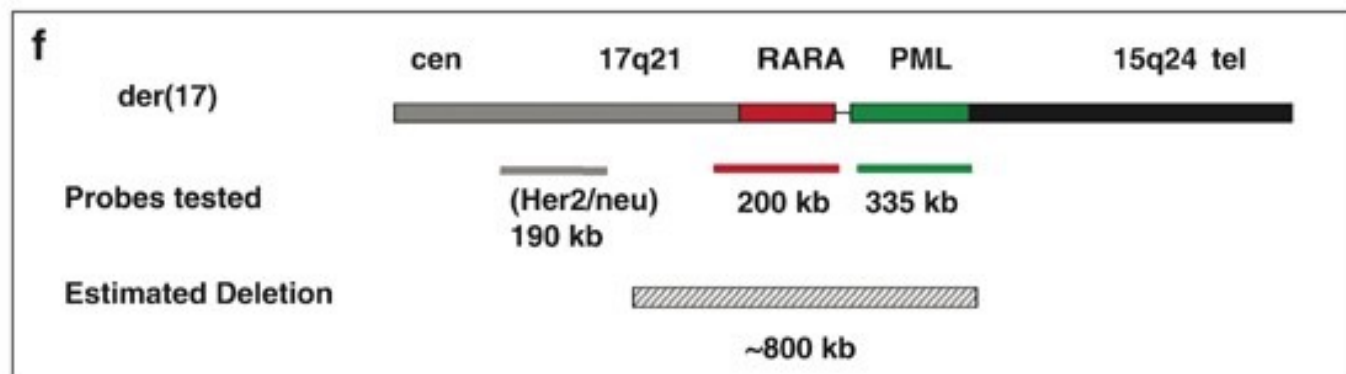
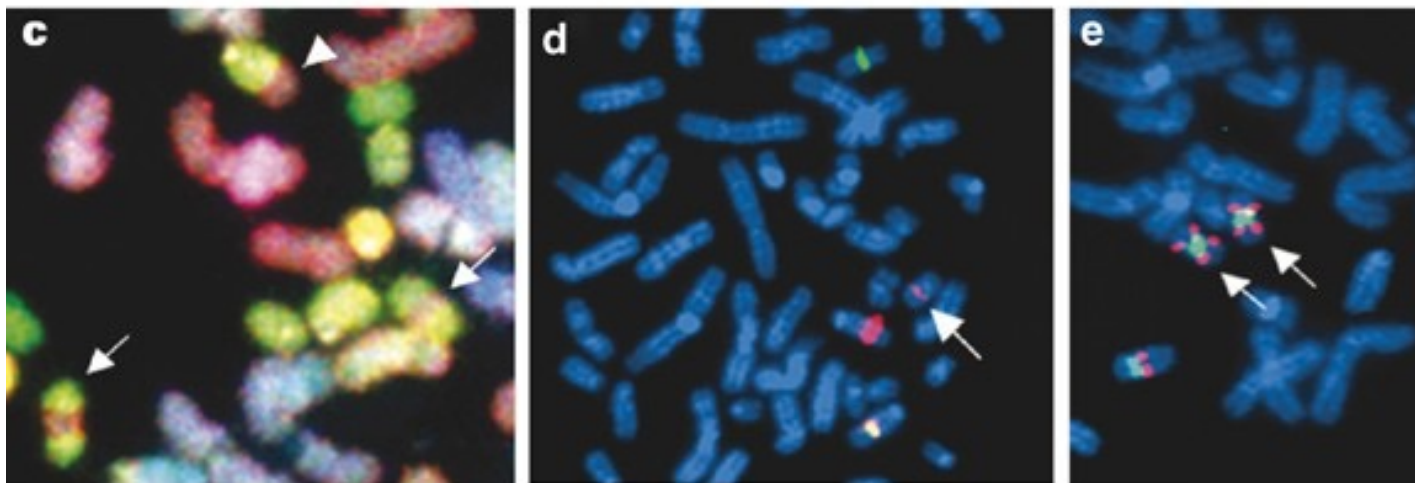
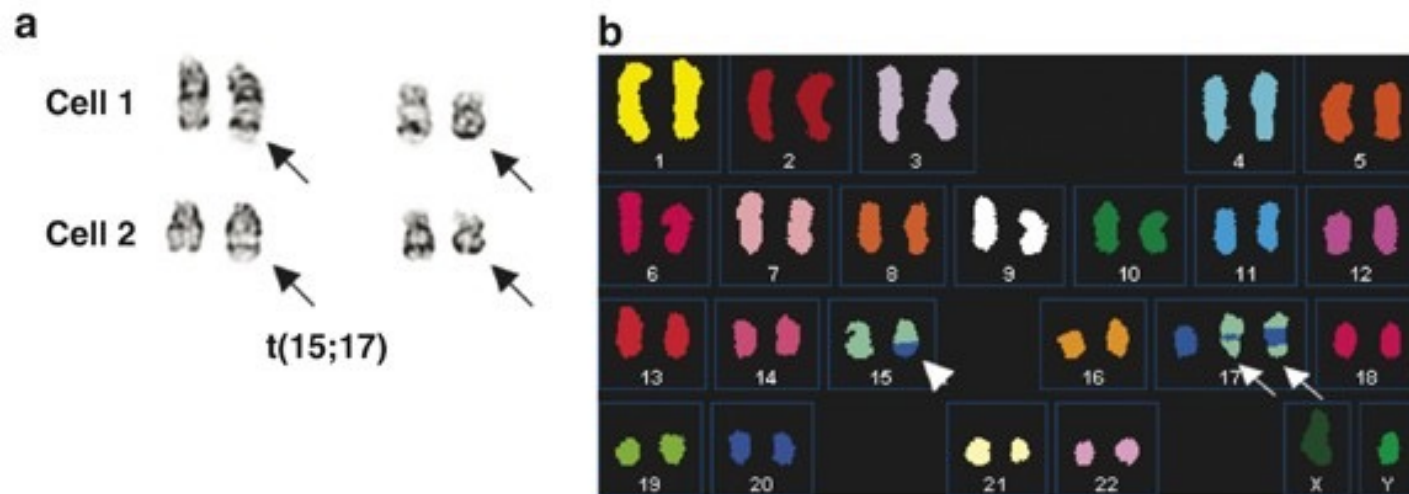


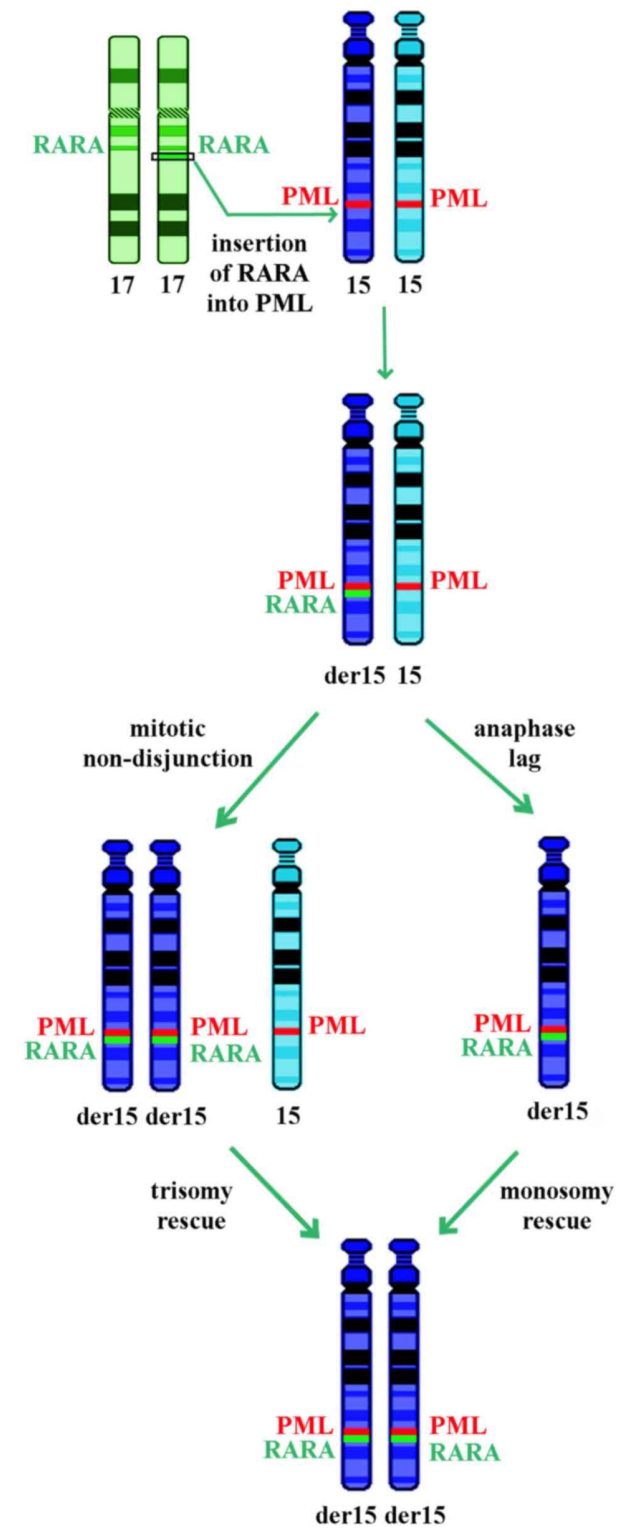
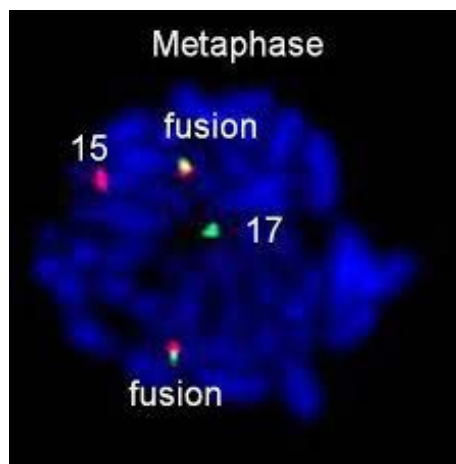
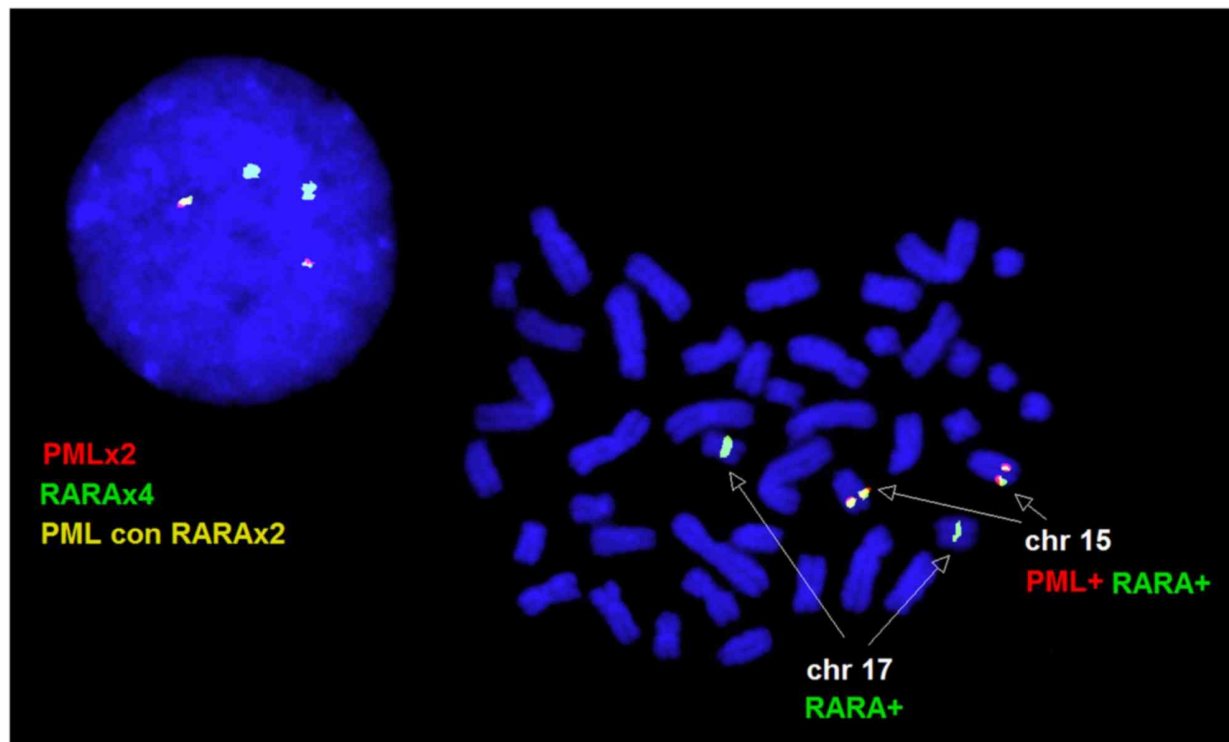
C





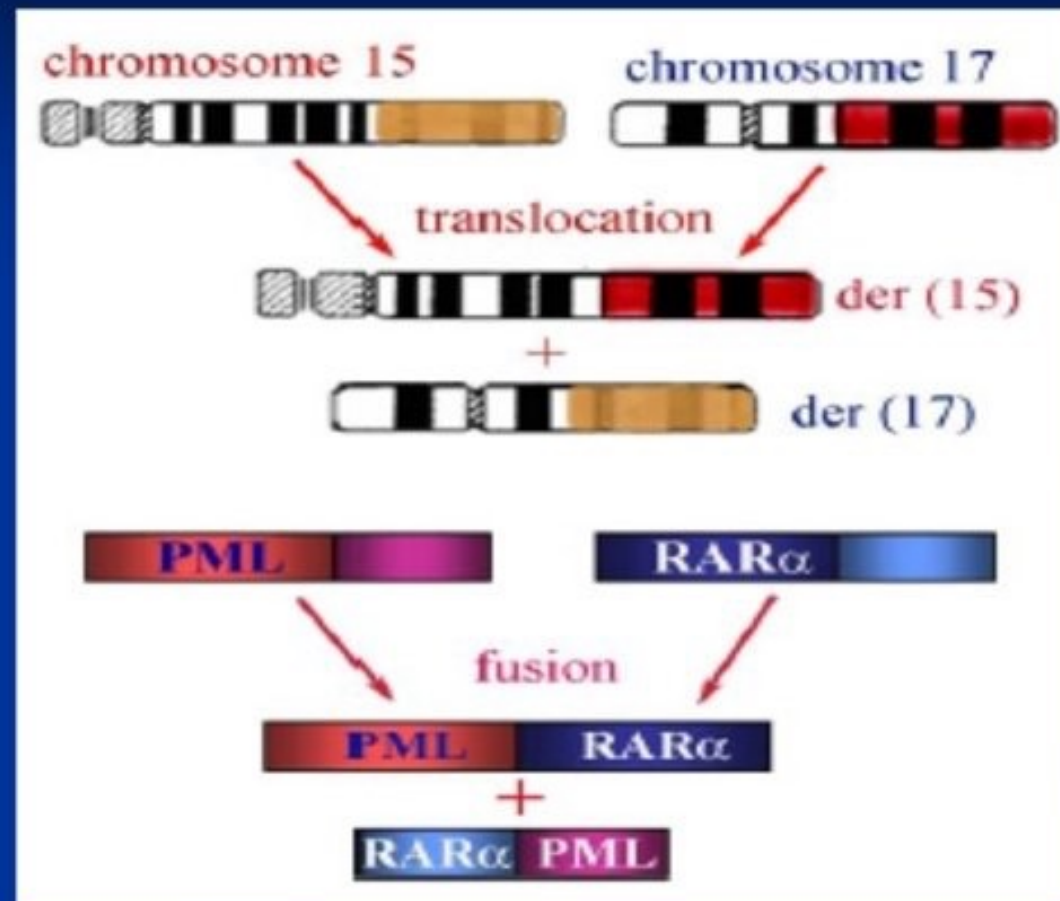


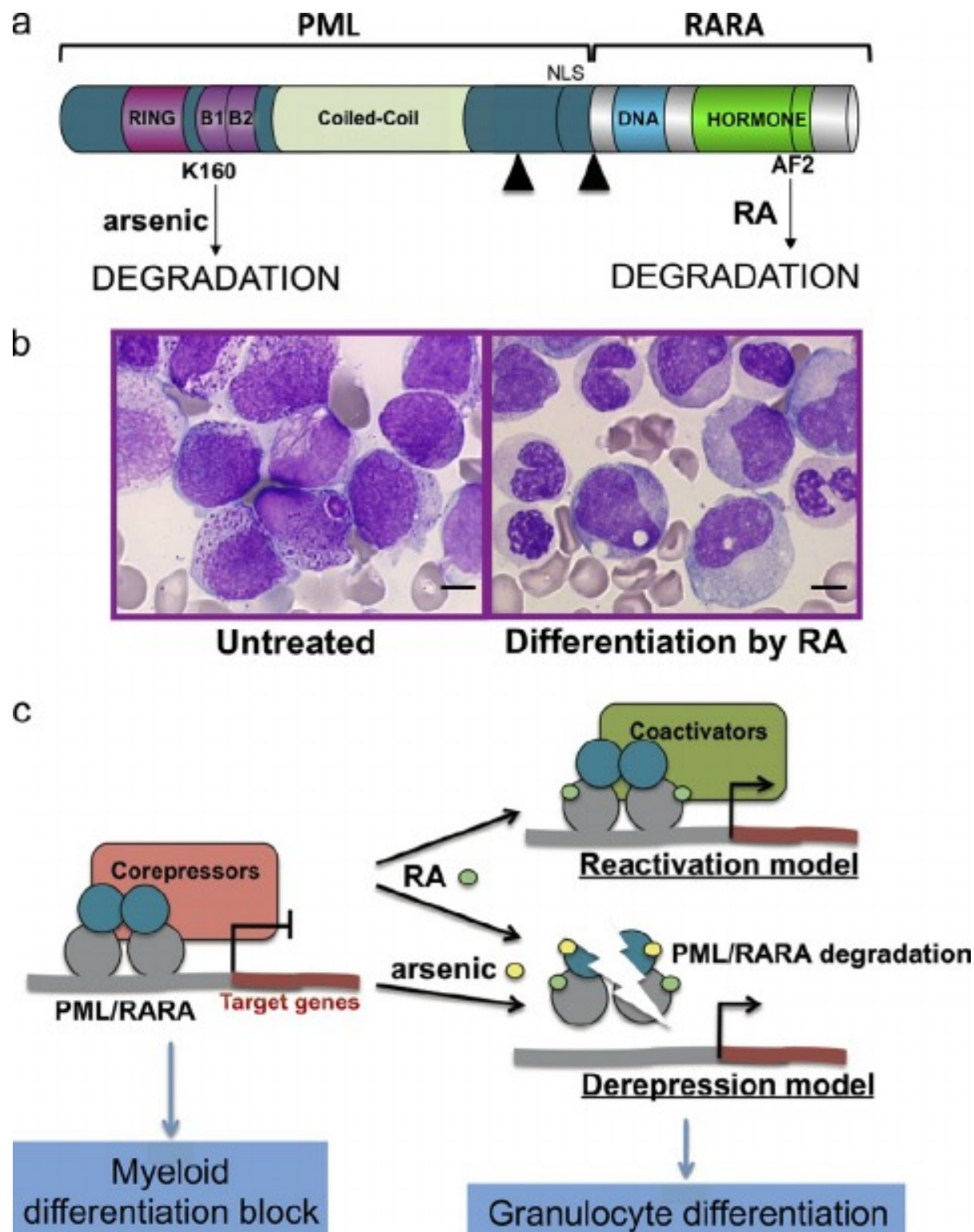


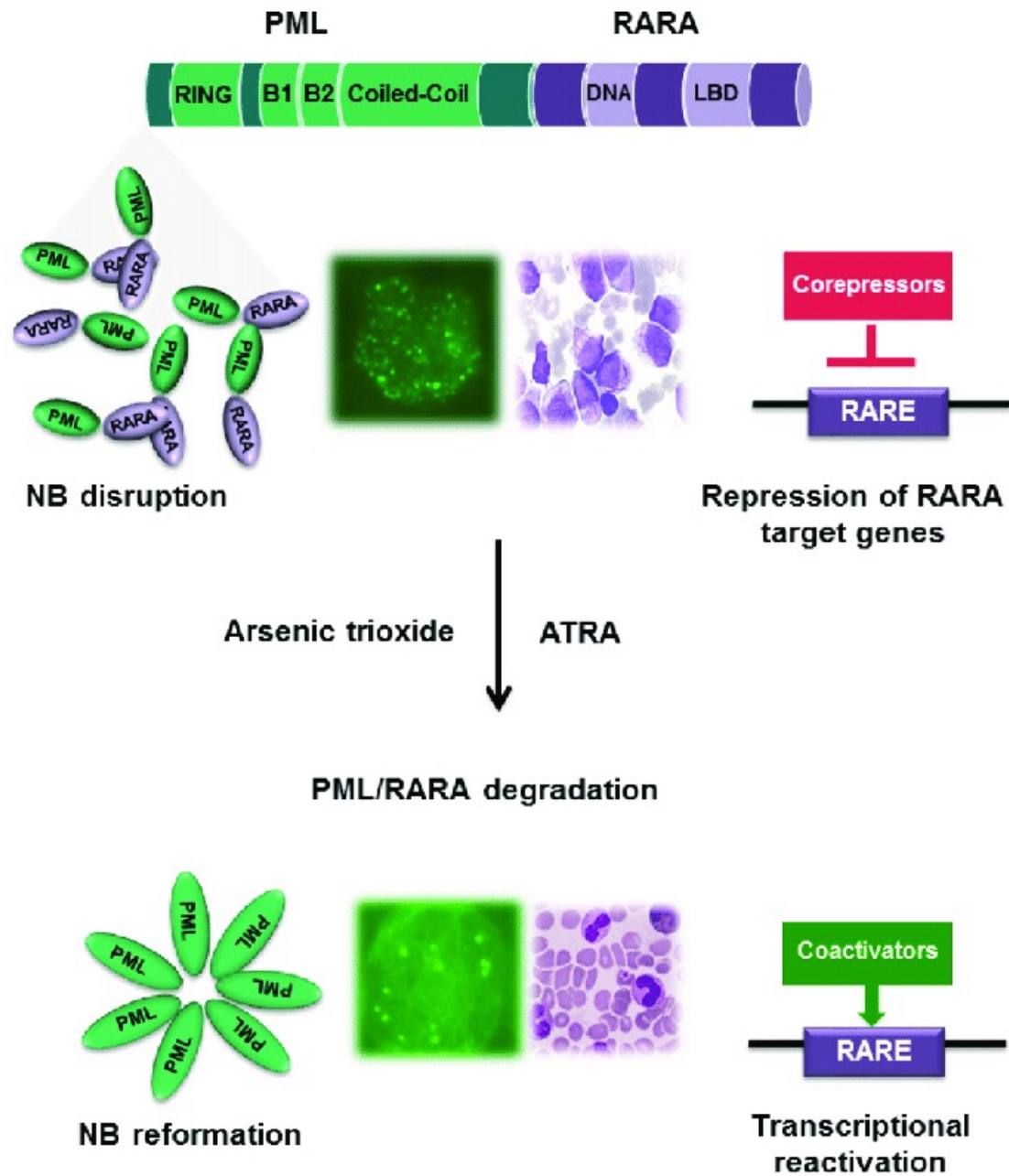


t(15;17); PML-RARA

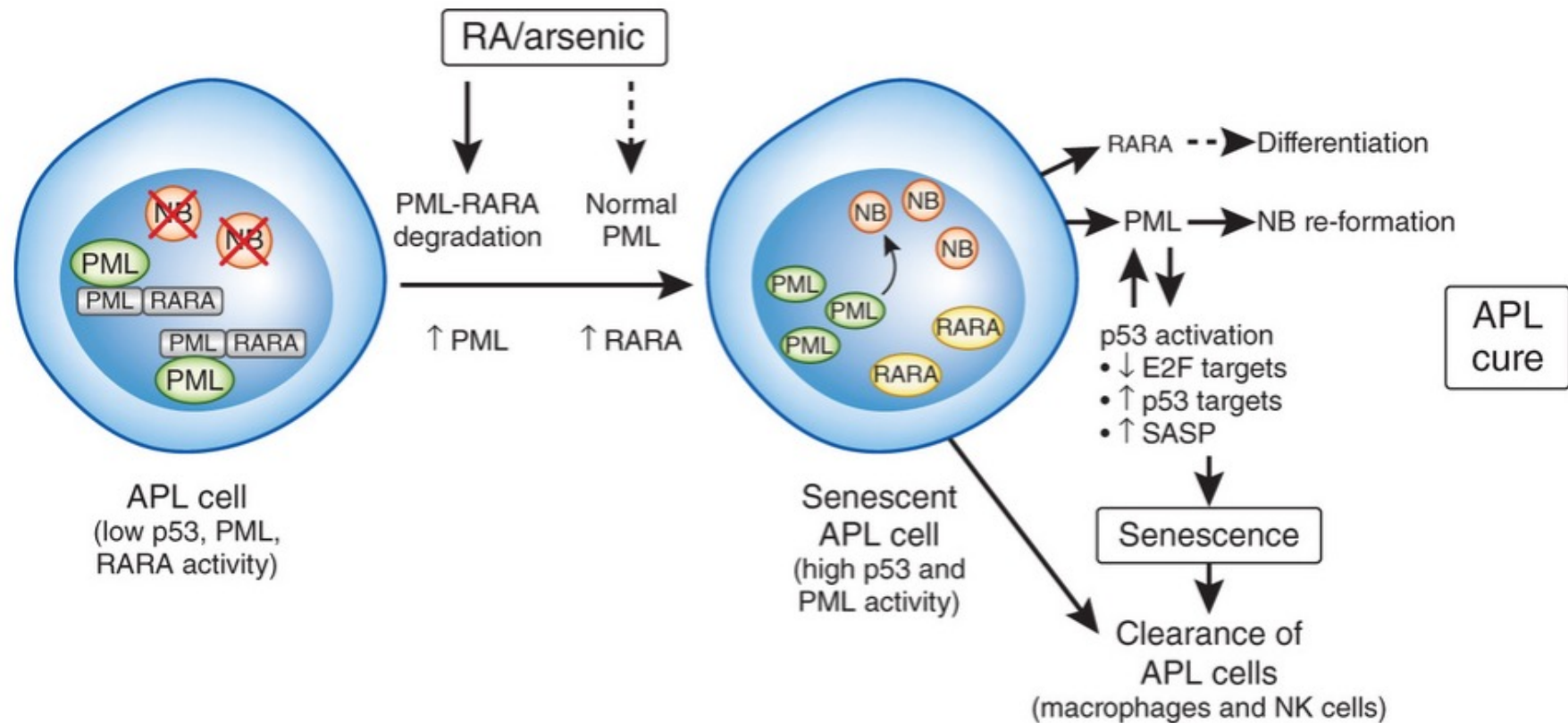
Physiologic quantities of retinoic acid no longer sufficient to allow for cell differentiation.







PML-p53 Axis Drives Cure in APL



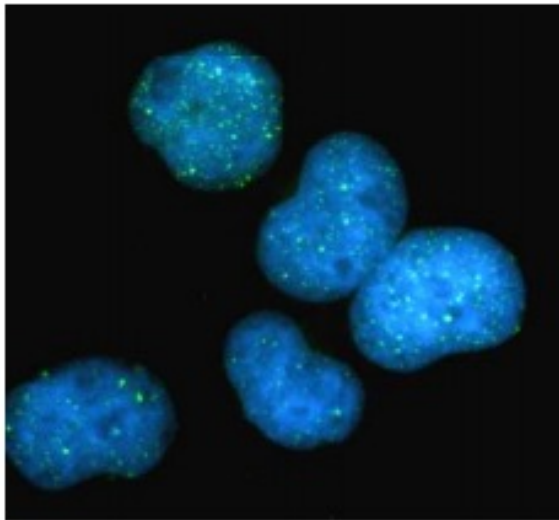
Nat. Med. 20(2):123-124 (2014)



PML-RARA in Leukemia

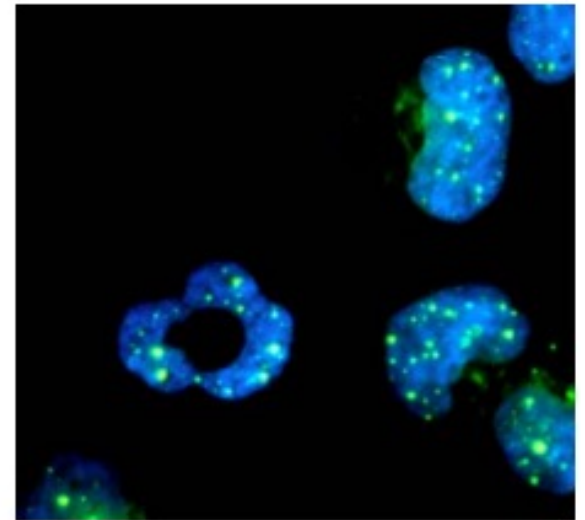
A

NB4

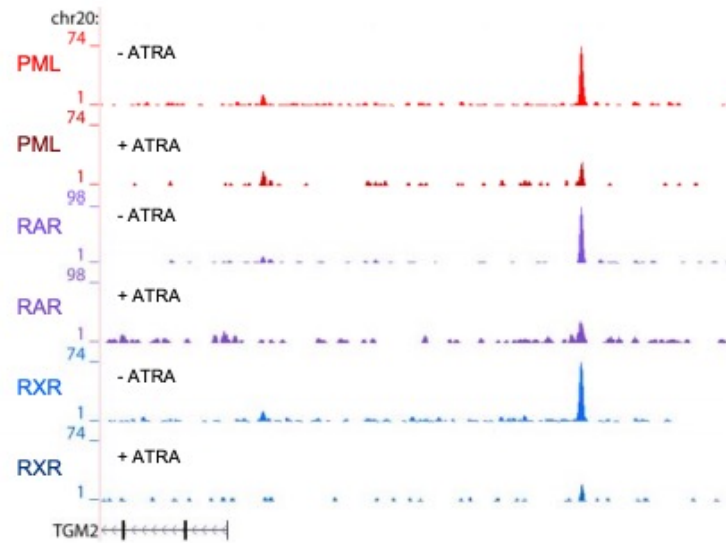


B

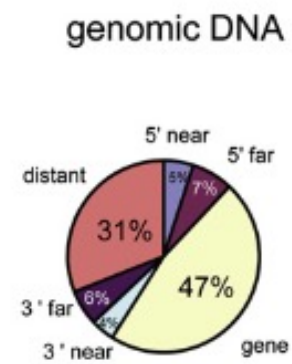
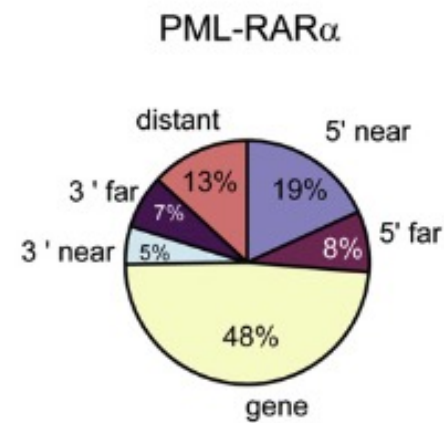
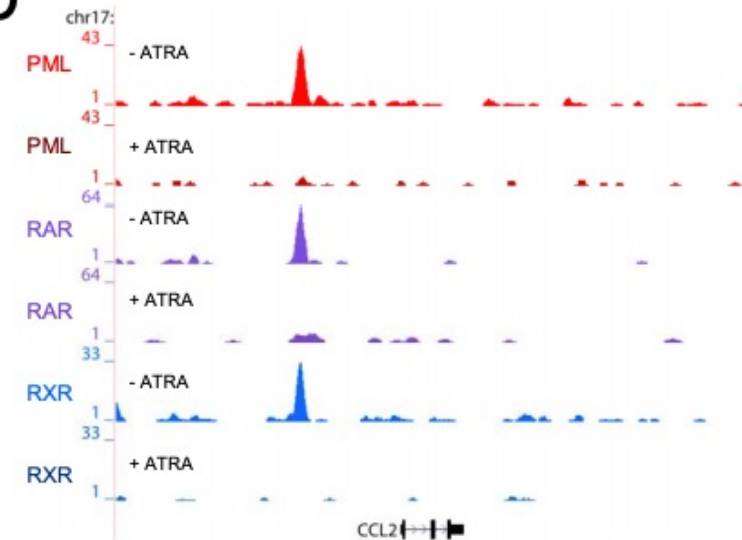
NB4 + ATRA

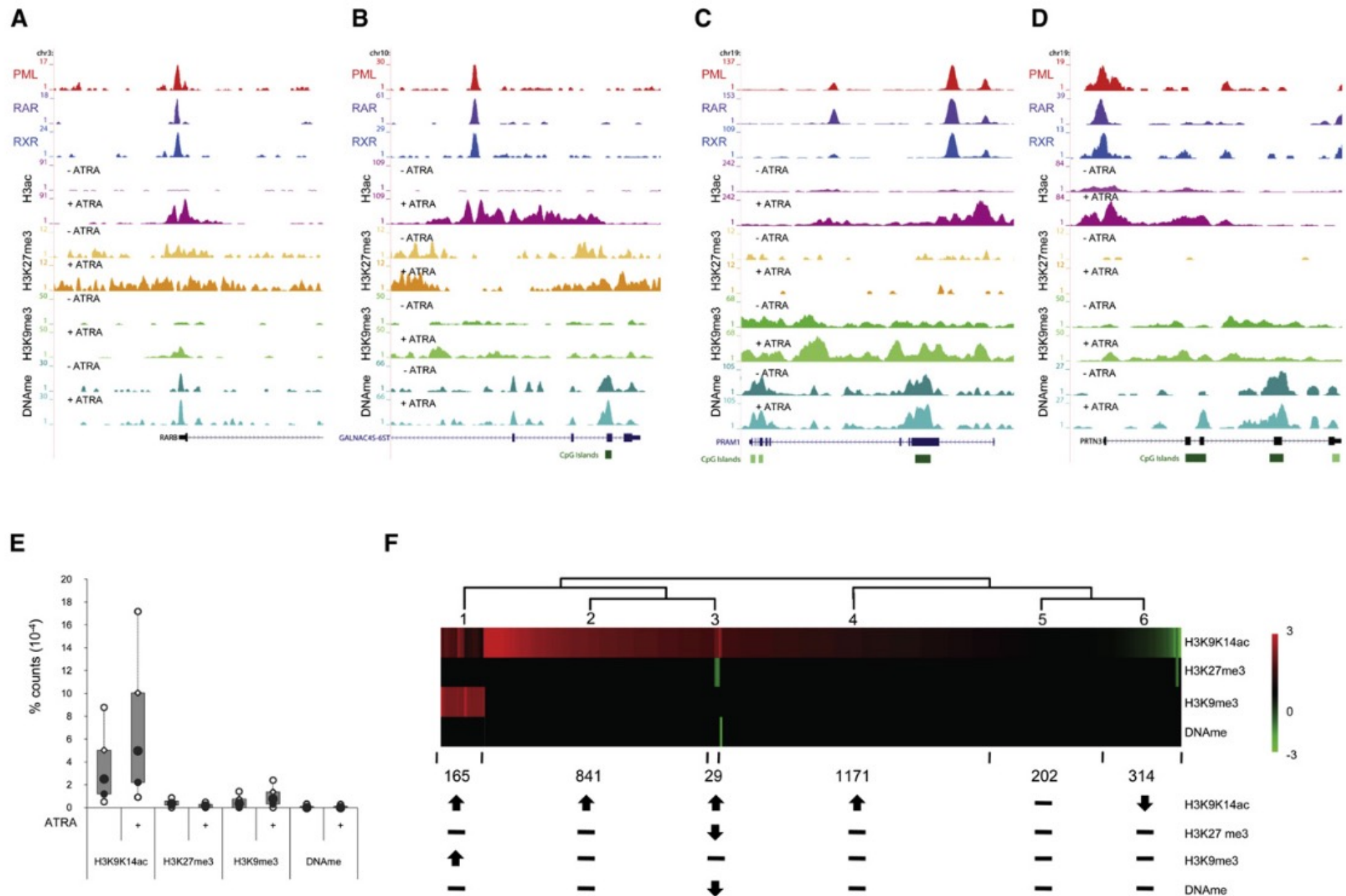


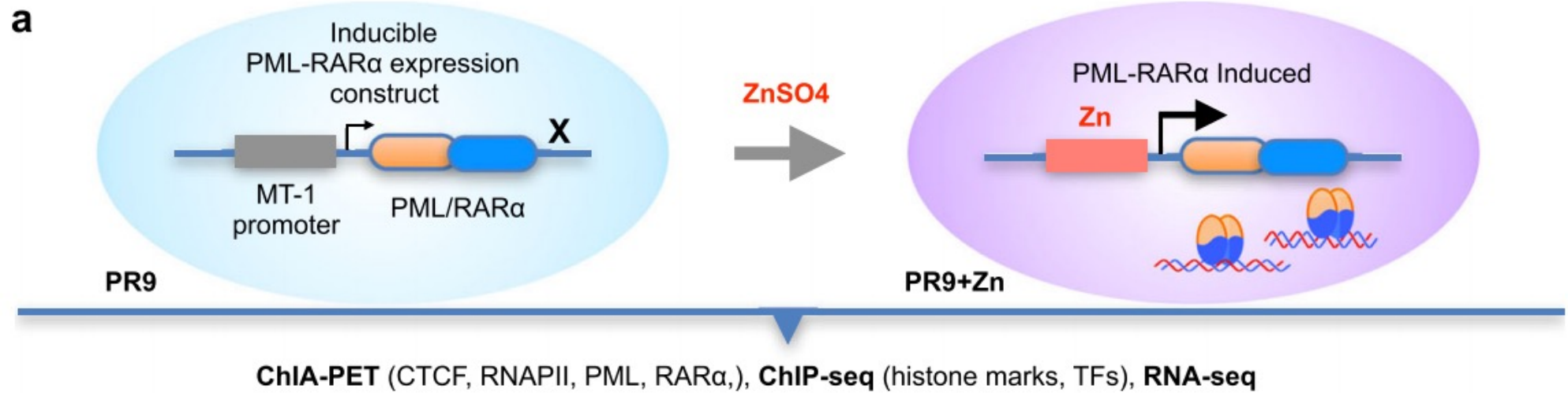
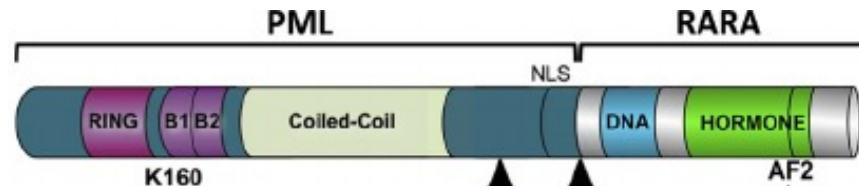
C

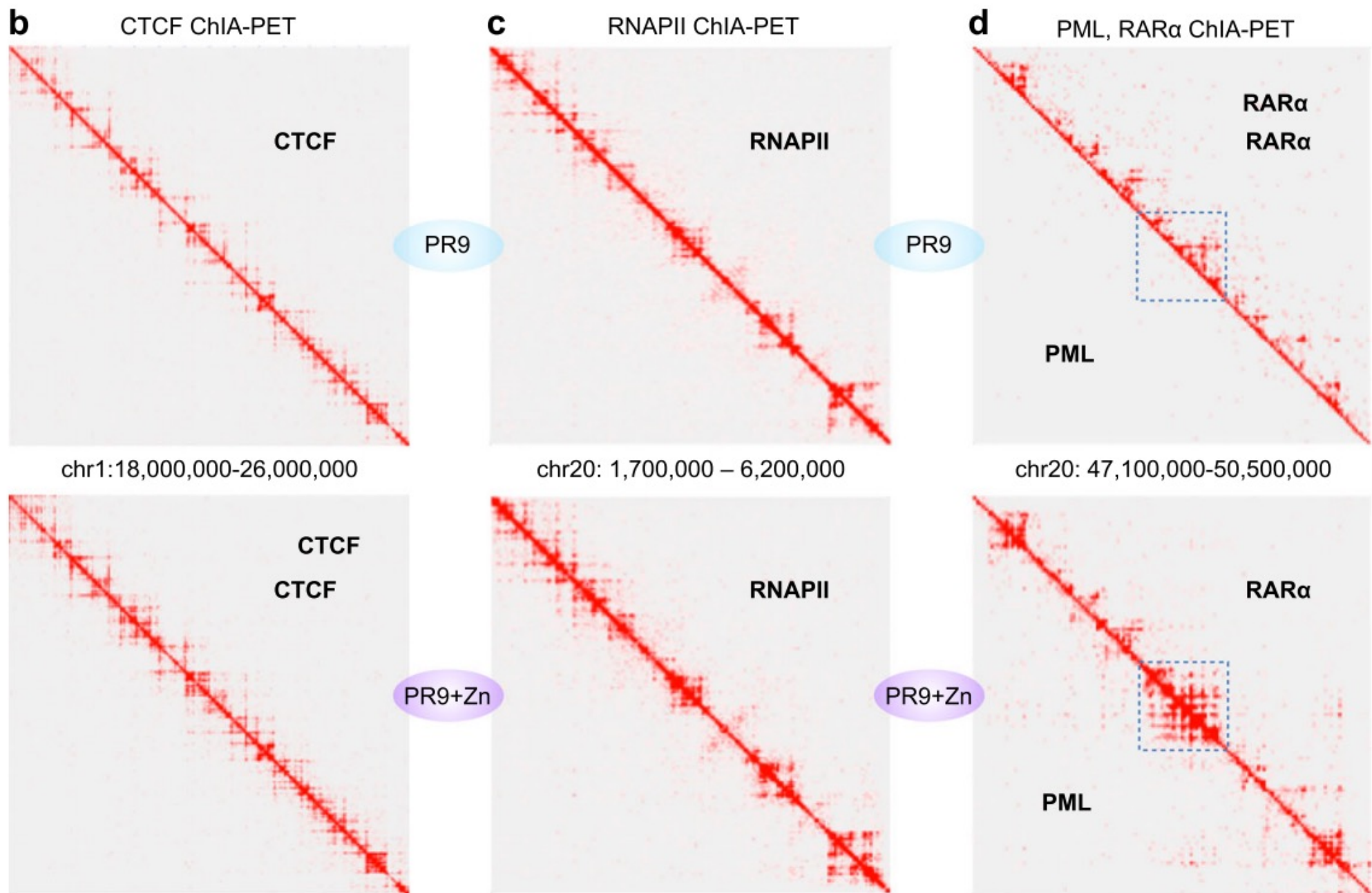


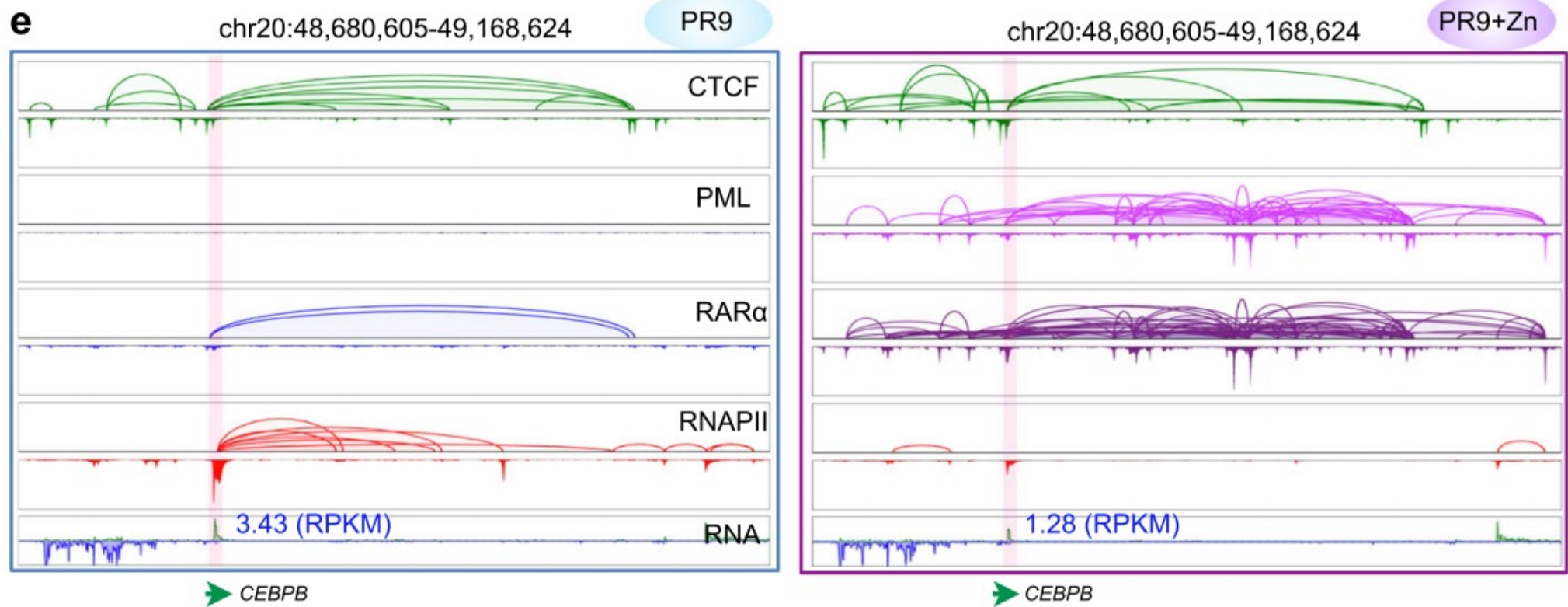
D

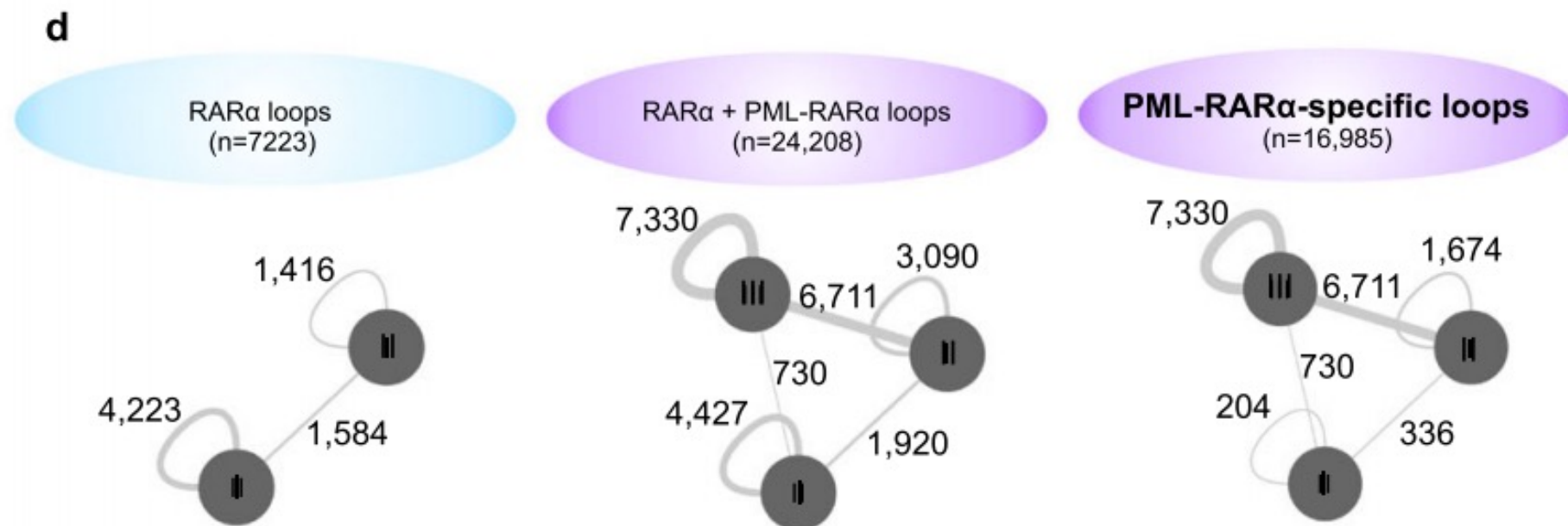
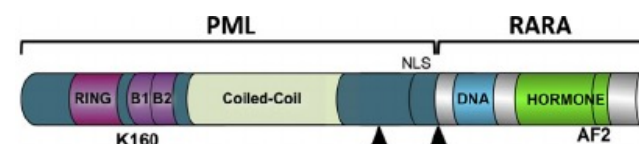
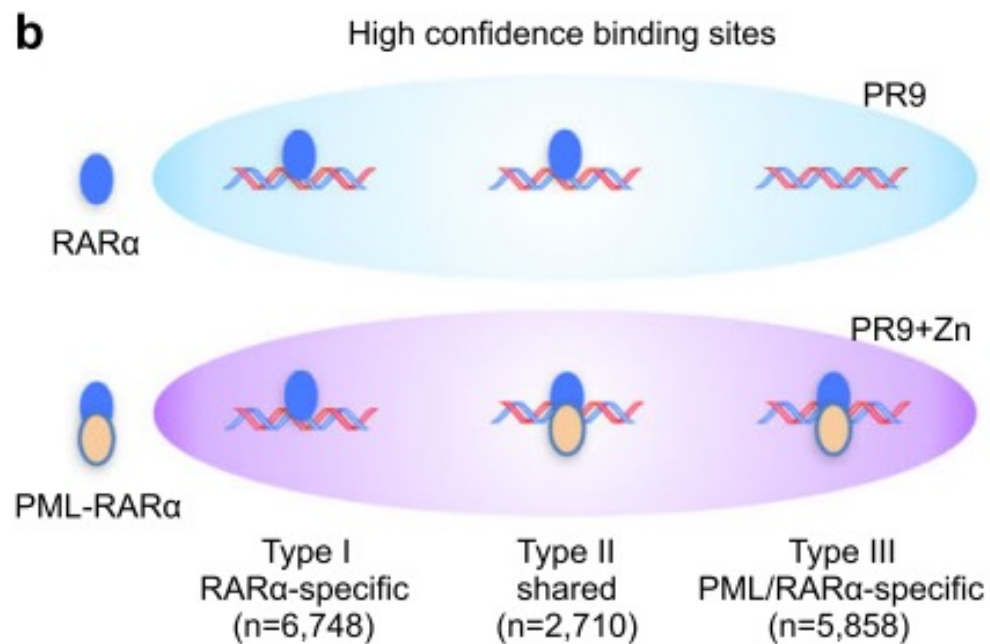


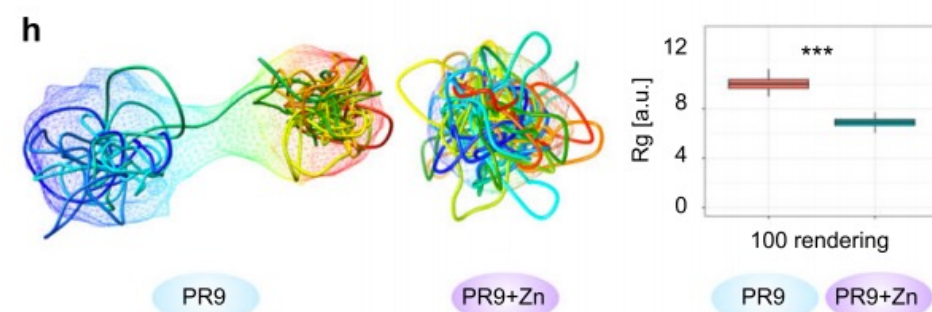
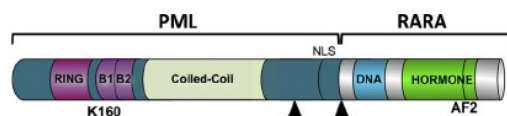
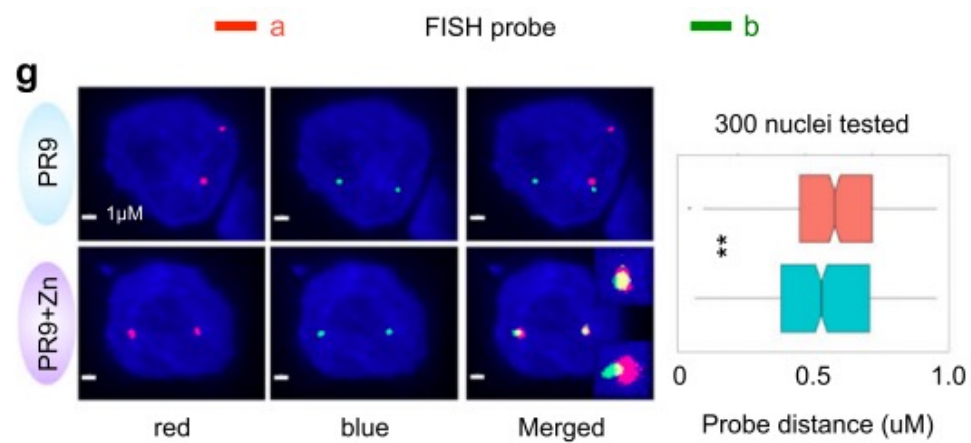
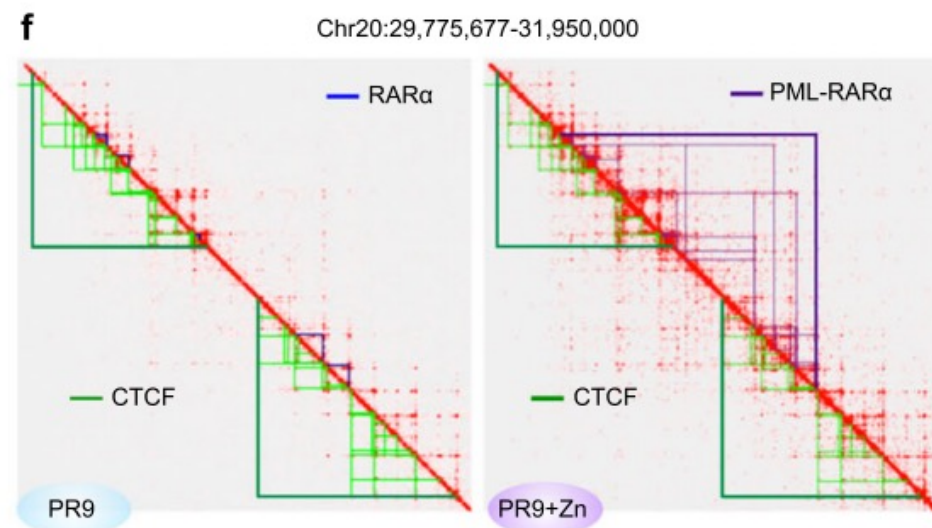
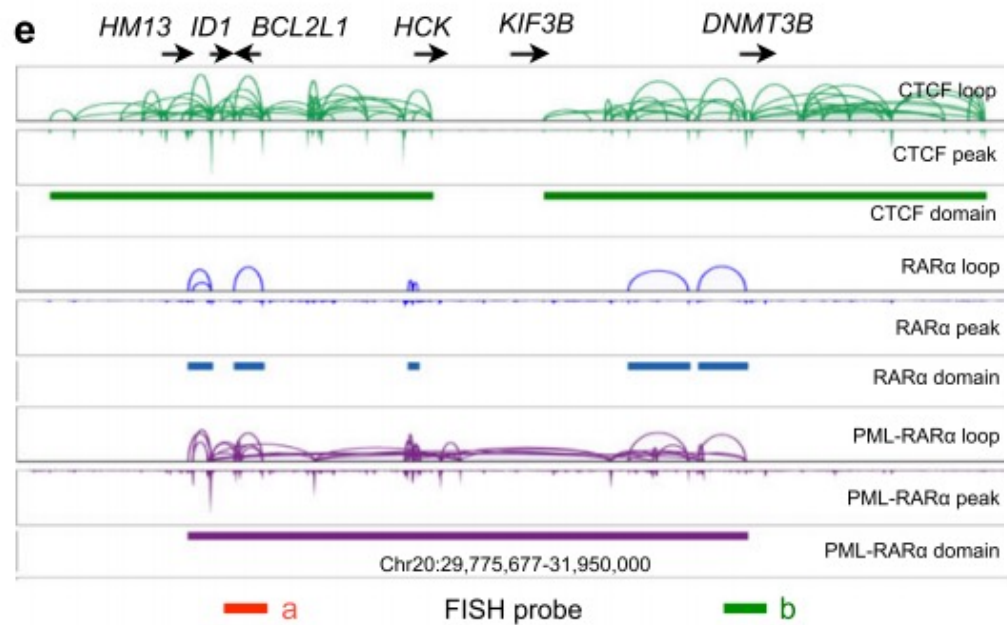


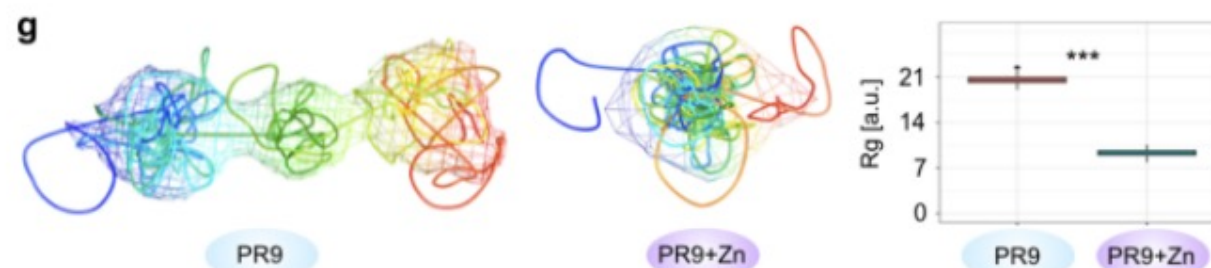
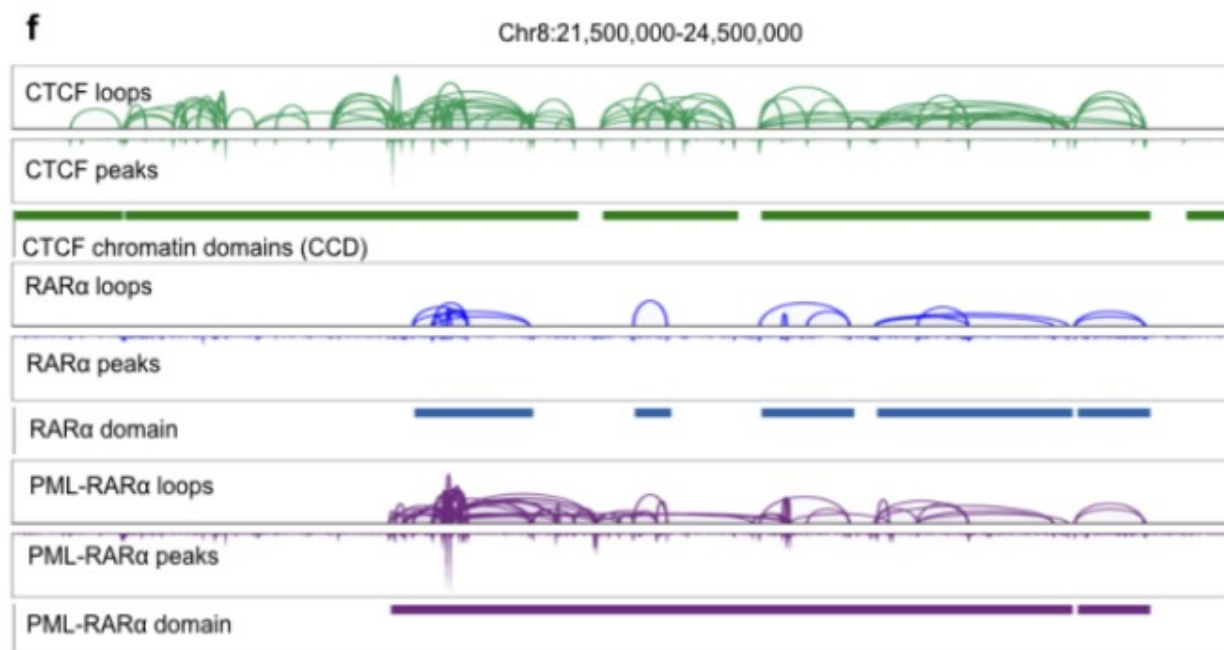
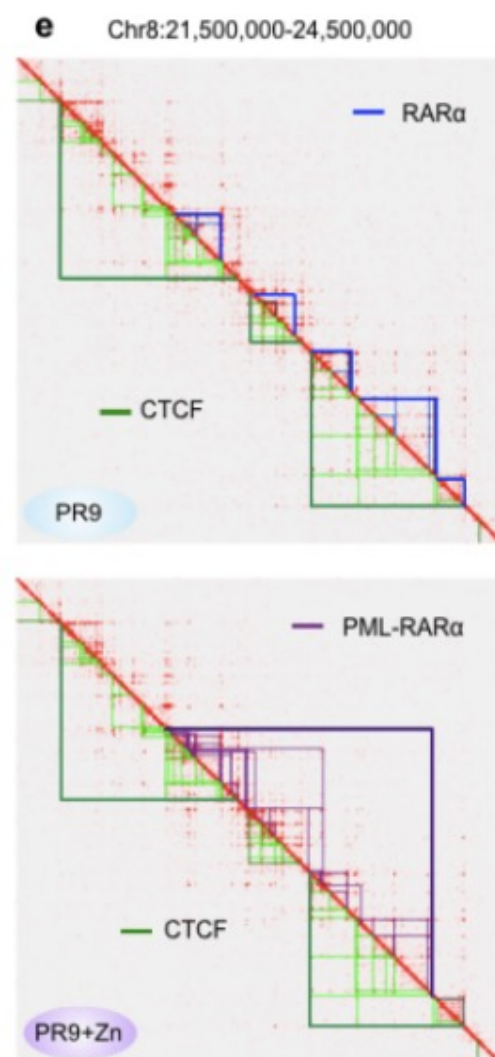


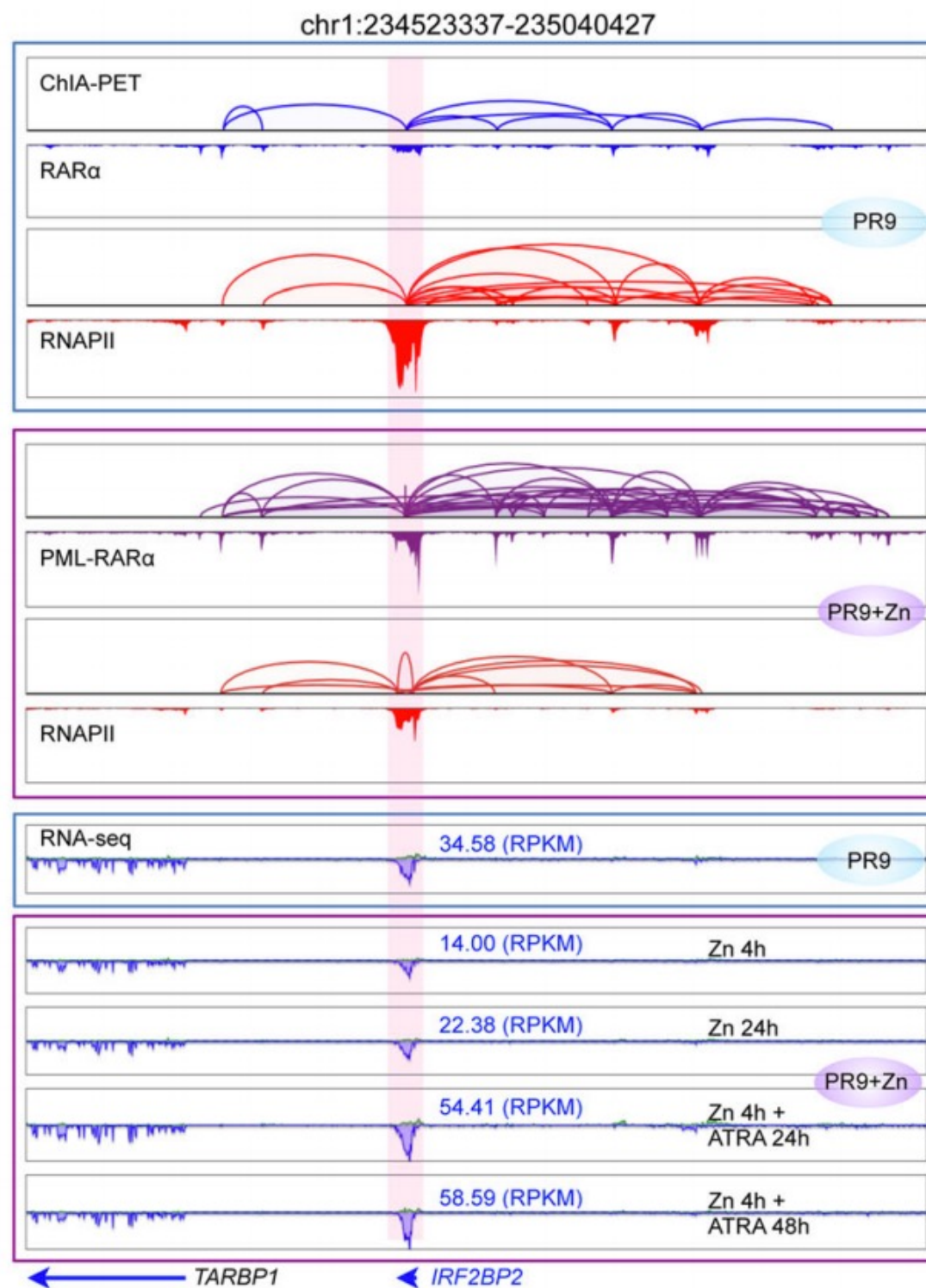


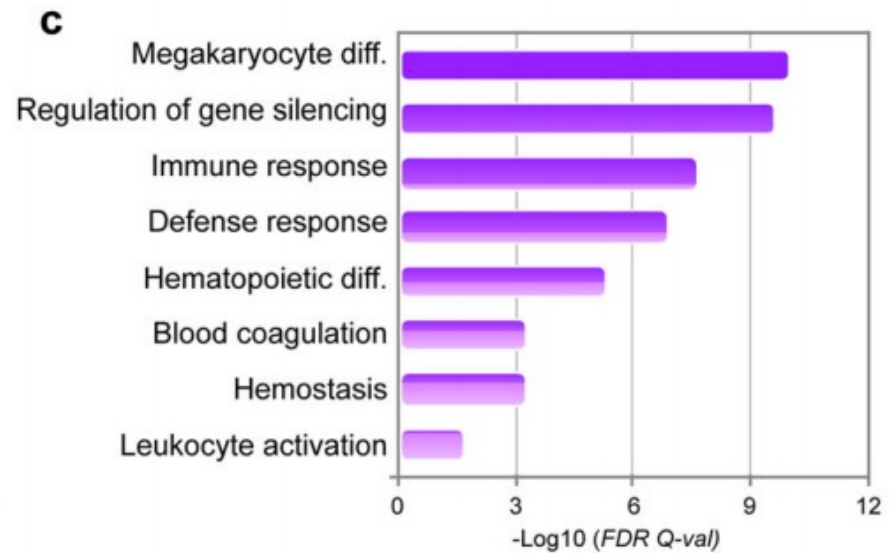
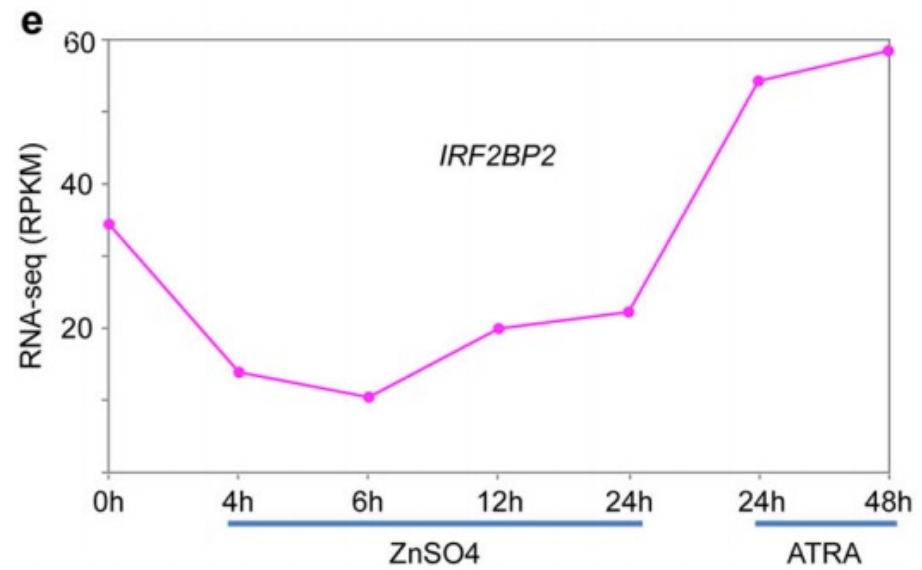
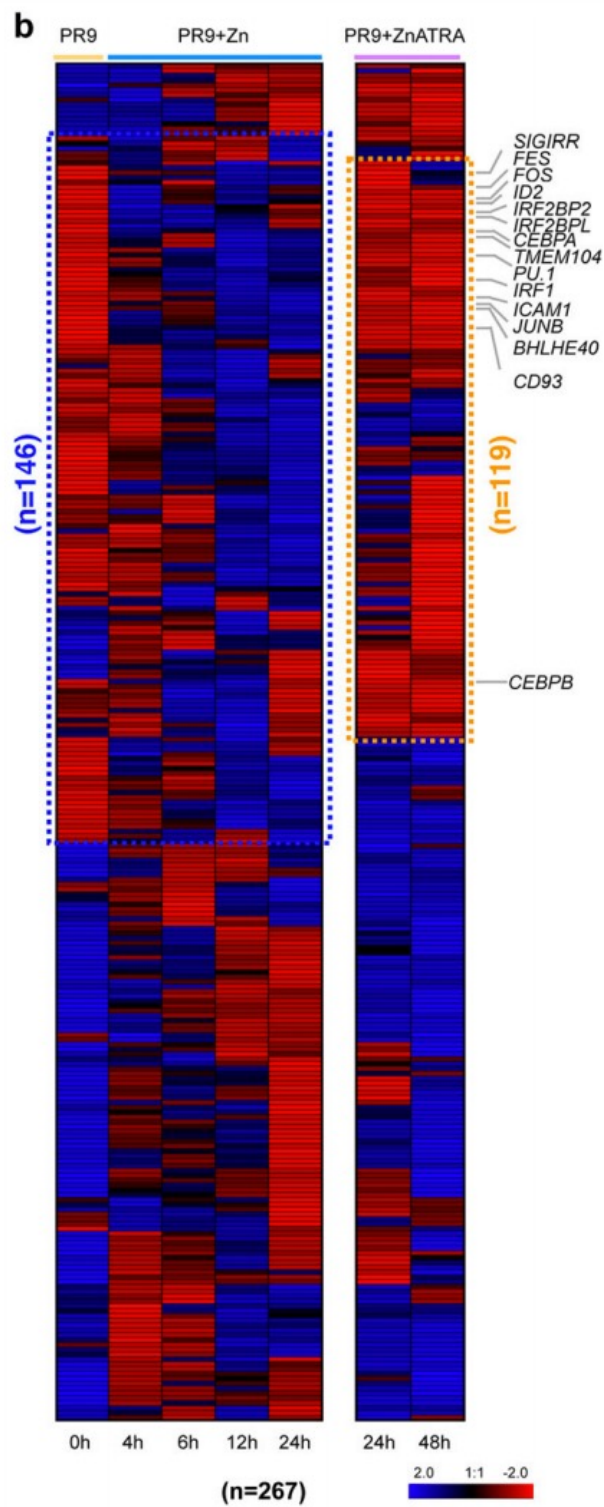


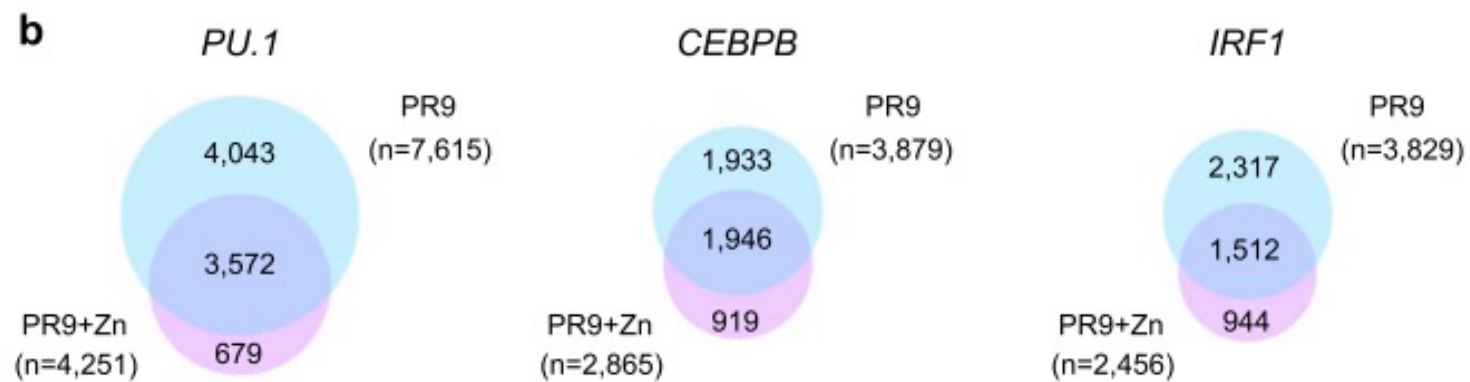
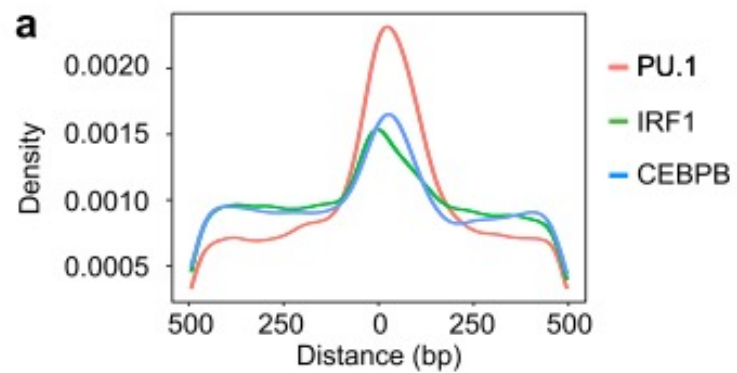






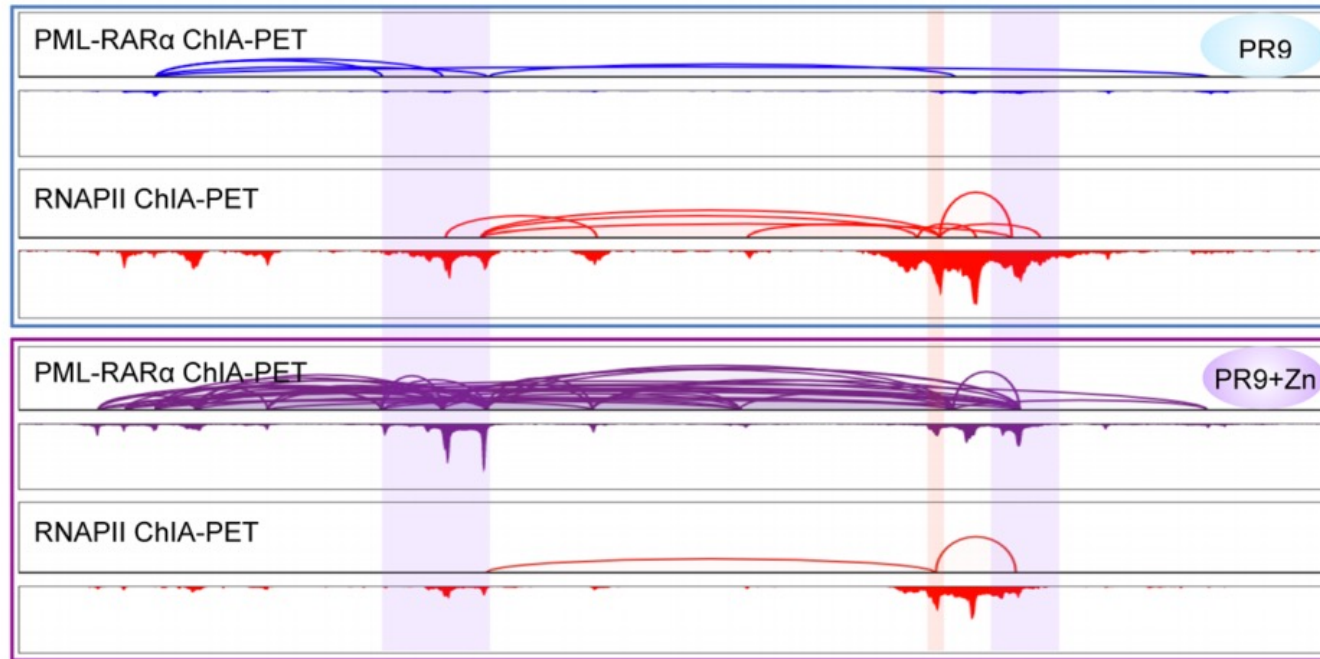
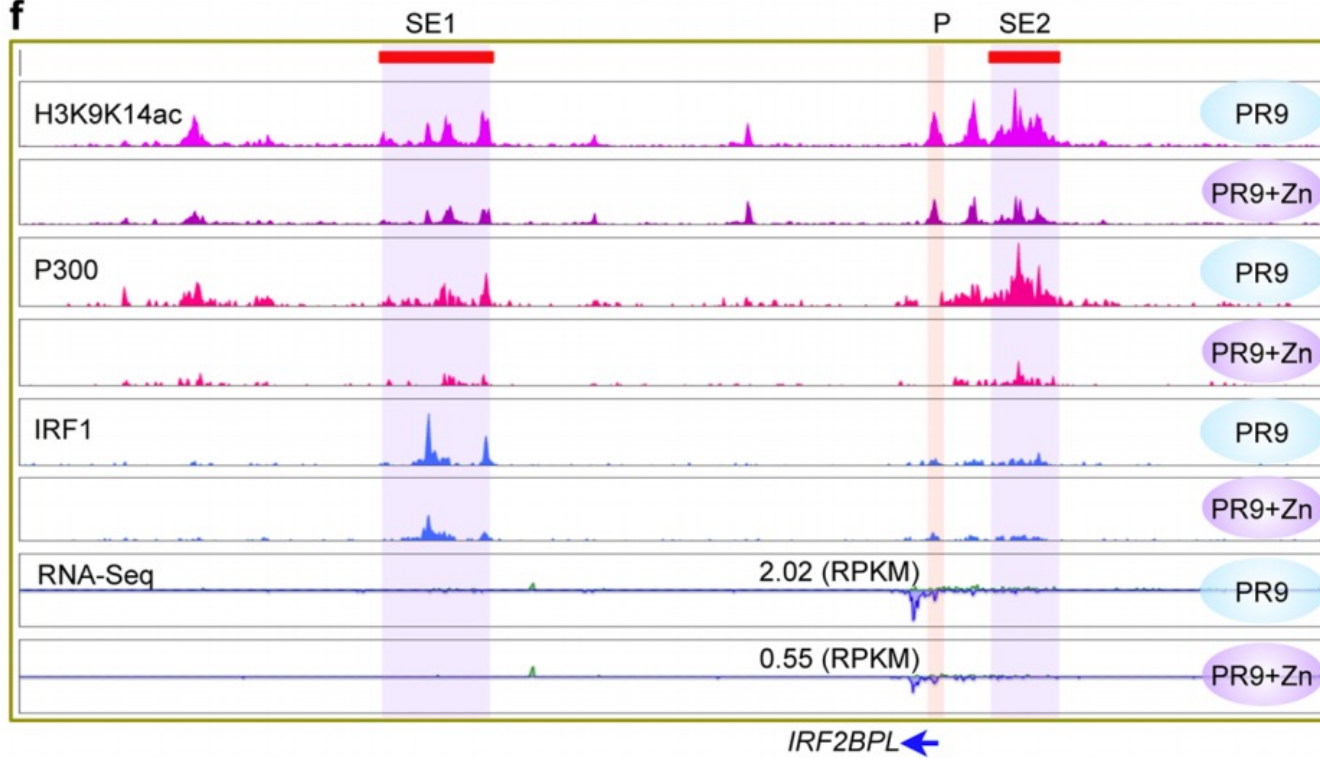


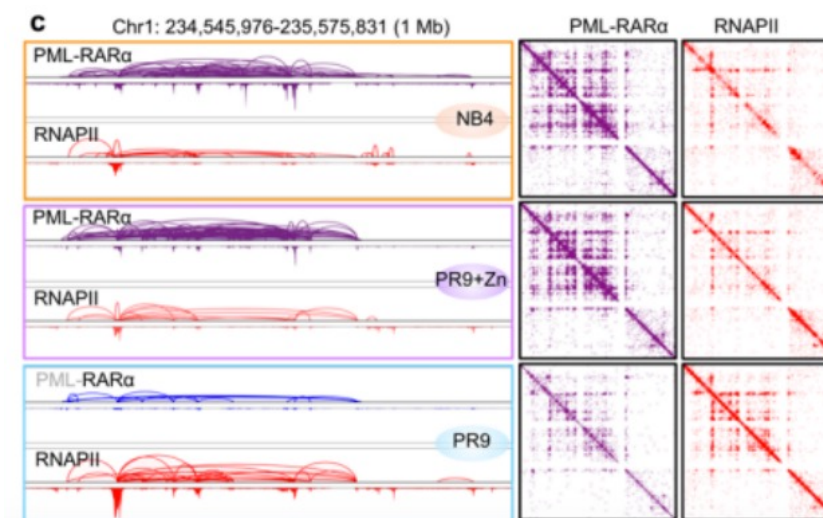
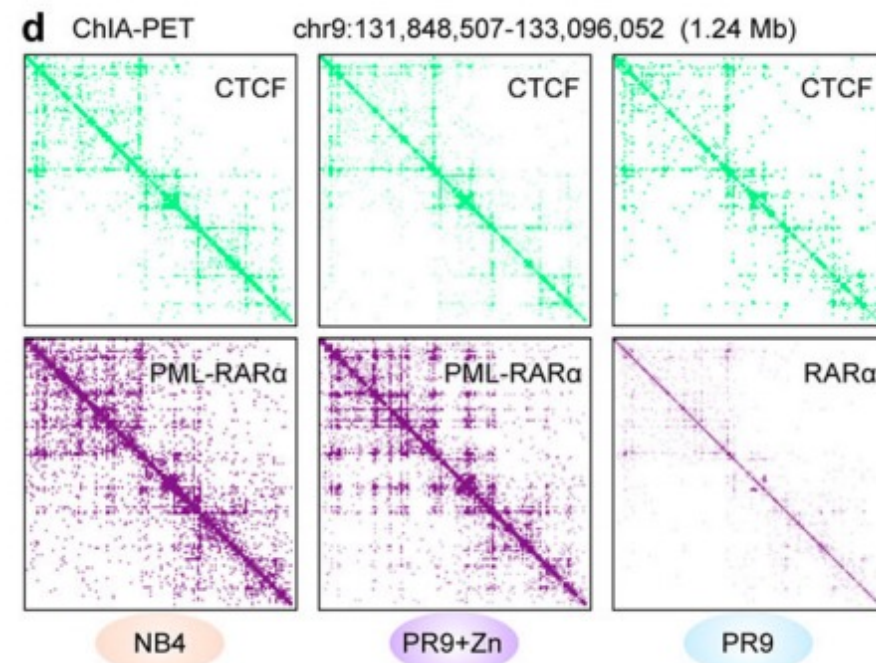
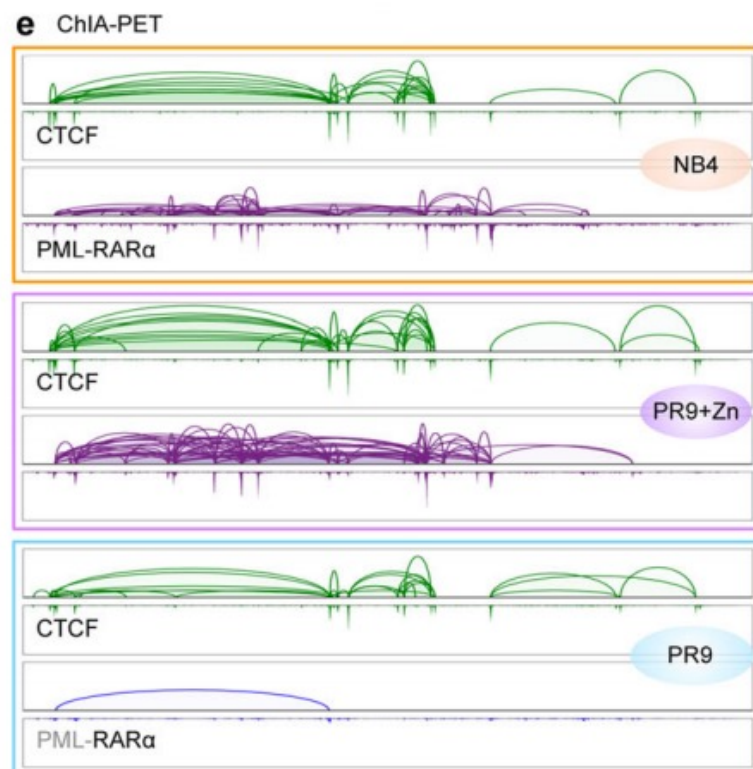
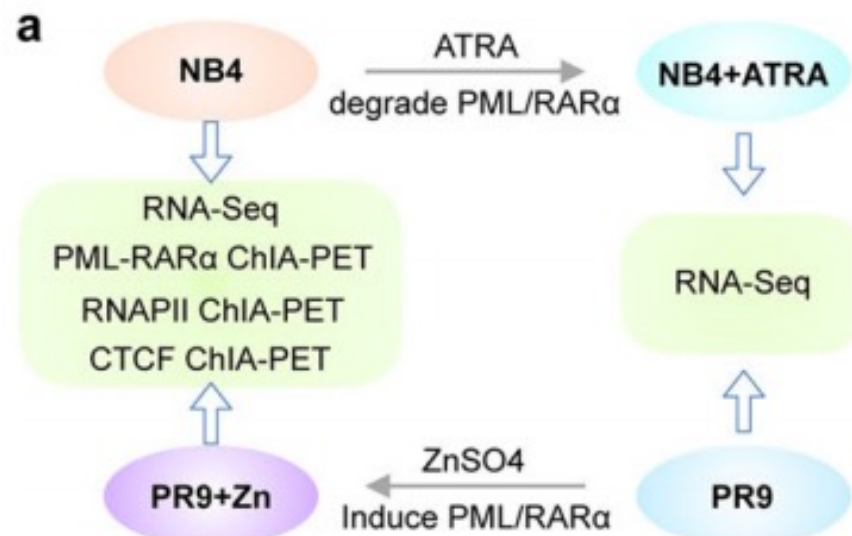


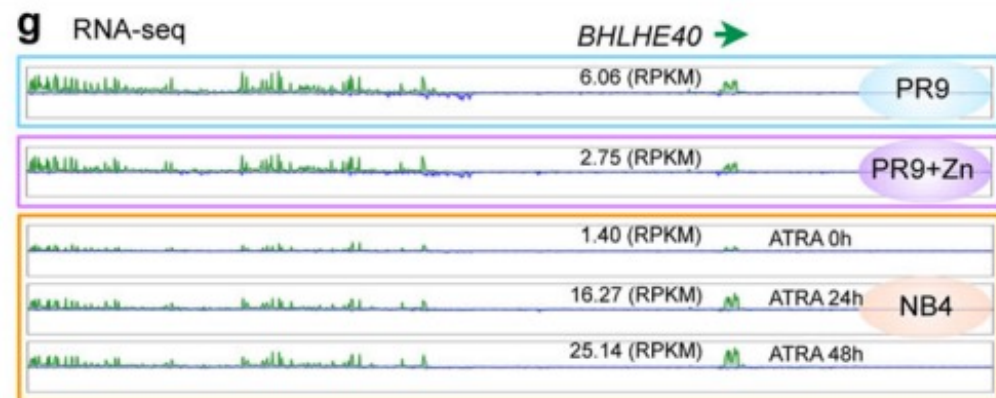
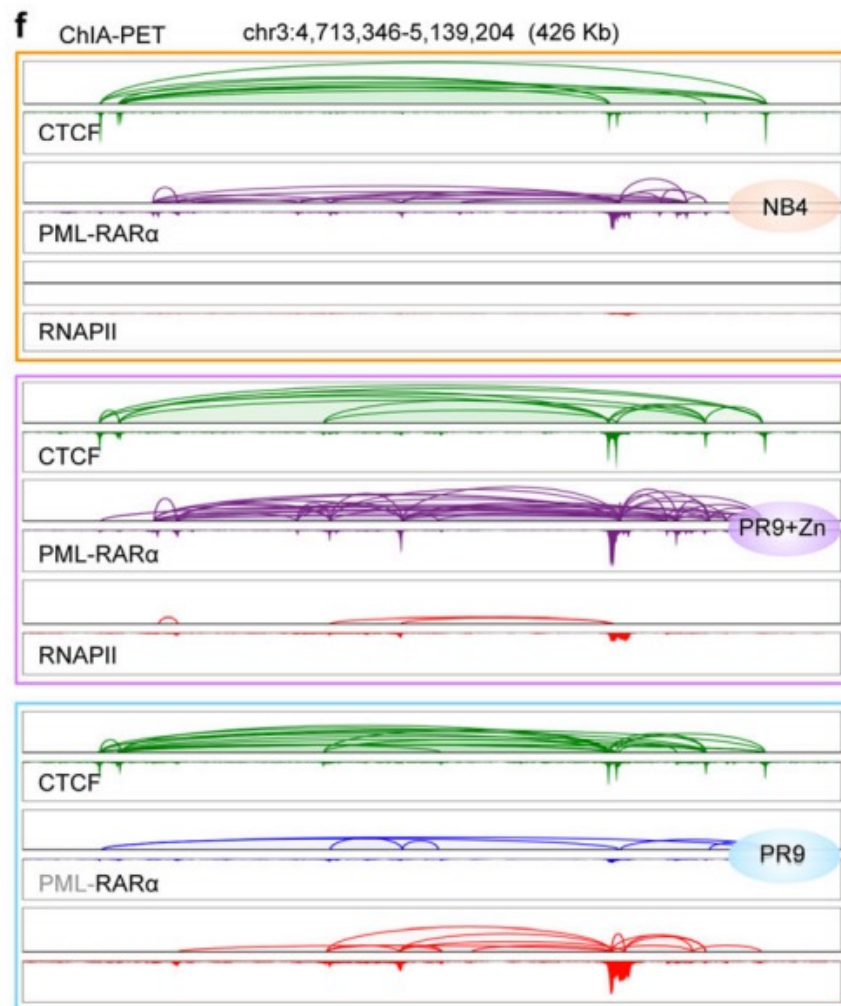


e

Chr14:77359915-77553224

**f**







Strategies for
diagnostic and
treatment
based on
Omics data